

Subspaces and Rank

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Rank

Definition 0.1 (Linearly independent vectors). Vectors $\tilde{v}_1, \dots, \tilde{v}_k$ of the same length p are **linearly independent** if the only numbers $c_1, \dots, c_k$ with

$$c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k = \tilde{0}$$

are

$$\begin{aligned} c_1 &= \dots \\ &= c_k \\ &= 0. \end{aligned}$$

Vectors that are not linearly independent are **linearly dependent**.

Example 0.1 (Independent and dependent pairs of vectors).

- $\tilde{v}_1 = (1, 0)$ and $\tilde{v}_2 = (1, 1)$ are linearly independent: $c_1 \tilde{v}_1 + c_2 \tilde{v}_2 = (c_1 + c_2, c_2)$, which is $\tilde{0}$ only if $c_2 = 0$ and then $c_1 = 0$.
- $\tilde{v}_1 = (1, 2)$ and $\tilde{v}_2 = (2, 4)$ are not: $2\tilde{v}_1 - \tilde{v}_2 = \tilde{0}$.

Definition 0.2 (Rank). The **rank** of a matrix $\mathbf{A}$, written $\text{rank}(\mathbf{A})$, is the largest number of columns of $\mathbf{A}$ that are linearly independent (Definition 0.1).

Example 0.2 (The rank of two 3×2 matrices). The matrix $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ has rank 2: if $c_1(1, 1, 1) + c_2(1, 2, 3) = \tilde{0}$, then subtracting the first entry from the second gives $c_2 = 0$,

and then $c_1 = 0$.

The matrix $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

Definition 0.3 (Full column rank). An $n \times p$ matrix has **full column rank** if its rank (Definition 0.2) is p , so that all of its columns are linearly independent.

Remark 0.1 (Full column rank needs at least as many rows as columns). In Example 0.2, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An $n \times p$ matrix can have full column rank only when $p \leq n$, because more than n vectors of length n are never linearly independent. For example, the 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ has 3 columns of length 2, and $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \tilde{0}$, with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

Subspaces

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Definition 0.4 (Subspace (linear subspace, vector space, linear space)). A set $\mathcal{S}$ of vectors in $\mathbb{R}^p$ is a **subspace** of $\mathbb{R}^p$ if it is not empty and it is closed under vector addition and scalar multiplication:

1. if $\tilde{u} \in \mathcal{S}$ and $\tilde{v} \in \mathcal{S}$, then $\tilde{u} + \tilde{v} \in \mathcal{S}$;
2. if $\tilde{u} \in \mathcal{S}$ and $c \in \mathbb{R}$, then $c\tilde{u} \in \mathcal{S}$.

A subspace is also called a **linear subspace**, a **vector space**, or a **linear space**.

Example 0.3 (A line through the origin is a subspace). Let $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$, the line in $\mathbb{R}^2$ through $(0, 0)$ and $(1, 2)$.

- **Addition:** $a(1, 2) + b(1, 2) = (a + b)(1, 2)$, which is in $\mathcal{S}$.
- **Scalar multiplication:** $c(a(1, 2)) = (ca)(1, 2)$, which is in $\mathcal{S}$.

So $\mathcal{S}$ is a subspace of $\mathbb{R}^2$. Two other subspaces of $\mathbb{R}^p$ are the set $\{\tilde{0}\}$ that holds only the zero vector, and $\mathbb{R}^p$ itself.

Example 0.4 (A line that misses the origin is not a subspace). Let $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$, the horizontal line in $\mathbb{R}^2$ at height 1. The vectors $(0, 1)$ and $(2, 1)$ are in $\mathcal{T}$, but their sum $(2, 2)$ is not, because its second entry is 2, not 1. So $\mathcal{T}$ is not closed under addition, and it is not a subspace.

Theorem 0.1 (Every subspace contains the zero vector). *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.4), then $\tilde{0} \in \mathcal{S}$.*

i Proof

Proof. A subspace is not empty, so it contains some vector $\tilde{u}$. Closure under scalar multiplication with $c = 0$ puts $0 \cdot \tilde{u}$ in $\mathcal{S}$, and $0 \cdot \tilde{u} = \tilde{0}$. $\square$

Example 0.5 (Using the zero vector to rule out a subspace). The plane $\{(x, y, z) : x + y + z = 1\}$ in $\mathbb{R}^3$ does not contain $\tilde{0} = (0, 0, 0)$, because $0 + 0 + 0 = 0 \neq 1$. So, by Theorem 0.1, it is not a subspace. The line $\mathcal{T}$ in Example 0.4 fails the same test:

$(0, 0)$ is not in $\mathcal{T}$, because its second entry is not 1.

Containing $\tilde{0}$ is necessary but not sufficient. The union of the two coordinate axes in $\mathbb{R}^2$, $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$, contains $\tilde{0}$, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis, so the union is not closed under addition and is not a subspace.

Definition 0.5 (Span). The **span** of vectors $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$ is the set of all their linear combinations:

$$\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\} \stackrel{\text{def}}{=} \{c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k : c_1, \dots, c_k \in \mathbb{R}\}.$$

The vectors $\tilde{v}_1, \dots, \tilde{v}_k$ **span** a set $\mathcal{S}$ if their span equals $\mathcal{S}$.

By convention, the span of the empty list ($k = 0$) is $\{\tilde{0}\}$, and the empty list counts as linearly independent (Definition 0.1), since it has no coefficients that could fail to be zero.

Example 0.6 (Two spans in $\mathbb{R}^3$).

- $\text{span}\{(1, 0, 0), (0, 1, 0)\}$ is the set of vectors $(c_1, c_2, 0)$, the plane $z = 0$ in $\mathbb{R}^3$.
- $\text{span}\{(1, 2, 3), (2, 4, 6)\}$ is only the line $\{c(1, 2, 3) : c \in \mathbb{R}\}$, because $(2, 4, 6) = 2(1, 2, 3)$, so $c_1(1, 2, 3) + c_2(2, 4, 6) = (c_1 + 2c_2)(1, 2, 3)$. Adding a vector to a list does not always enlarge its span.

Theorem 0.2 (A span is a subspace). *For any vectors $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$, $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$ (Definition 0.5) is a subspace of $\mathbb{R}^p$ (Definition 0.4).*

Proof

Proof. The span is not empty: it contains $0 \cdot \tilde{v}_1 + \dots + 0 \cdot \tilde{v}_k = \tilde{0}$.

Addition. Take two vectors in the span, $\tilde{u} = a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k$ and $\tilde{w} = b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k$. Then

$$\begin{aligned} \tilde{u} + \tilde{w} &= (a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) + (b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k) && \text{(substitute the two combinations)} \\ &= (a_1 + b_1) \tilde{v}_1 + \dots + (a_k + b_k) \tilde{v}_k && \text{(regroup, and distribute each } \tilde{v}_i \text{)} \end{aligned}$$

which is a linear combination of $\tilde{v}_1, \dots, \tilde{v}_k$, so it is in the span.

Scalar multiplication. For $c \in \mathbb{R}$,

$$\begin{aligned} c \tilde{u} &= c(a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) && \text{(substitute the combination)} \\ &= (ca_1) \tilde{v}_1 + \dots + (ca_k) \tilde{v}_k && \text{(distribute } c \text{ over the sum)} \end{aligned}$$

which is again in the span. □

Example 0.7 (The line in Example 0.3 is a span). The line $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$ in Example 0.3 is $\text{span}\{(1, 2)\}$, so Theorem 0.2 gives a second proof that it is a subspace. Likewise, the plane $z = 0$ in Example 0.6 is a subspace of $\mathbb{R}^3$, because it is $\text{span}\{(1, 0, 0), (0, 1, 0)\}$.

Theorem 0.3 (A matrix-vector product combines the columns). *If $\mathbf{A}$ is an $m \times n$ matrix with columns $\tilde{a}_1, \dots, \tilde{a}_n$ and $\tilde{x} \in \mathbb{R}^n$, then*

$$\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n.$$

i Proof

Proof. Compare entry i of the two sides, for each $i = 1, \dots, m$. Entry i of $\tilde{a}_j$ is a_{ij} , so

$$\begin{aligned} (\mathbf{A}\tilde{x})_i &= a_{i1}x_1 + \dots + a_{in}x_n && \text{(Definition 11 in Matrices)} \\ &= x_1a_{i1} + \dots + x_na_{in} && \text{(multiplication of numbers is commutative)} \\ &= (x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n)_i && \text{(entry } i \text{ of each } x_j\tilde{a}_j \text{ is } x_ja_{ij}) \end{aligned}$$

□

Example 0.8 (A product as a combination of columns).

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= 1 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} && \text{(Theorem 0.3)} \\ &= \begin{bmatrix} -1 \\ -1 \end{bmatrix}, && \text{(arithmetic)} \end{aligned}$$

the same answer as computing each entry as a row-times-vector dot product (Remark 3 in Matrices).

Theorem 0.4 (A matrix maps a combination to the same combination of images). *If $\mathbf{A}$ is an $m \times n$ matrix, $\tilde{z}_1, \dots, \tilde{z}_k \in \mathbb{R}^n$ and $c_1, \dots, c_k \in \mathbb{R}$, then*

$$\mathbf{A}(c_1\tilde{z}_1 + \dots + c_k\tilde{z}_k) = c_1\mathbf{A}\tilde{z}_1 + \dots + c_k\mathbf{A}\tilde{z}_k.$$

i Proof

Proof. The function $\tilde{x} \mapsto \mathbf{A}\tilde{x}$ is a linear map by Theorem 9 in Matrices, and a linear map preserves linear combinations by Theorem 10 in Matrices. Applying that theorem with $f(\tilde{x}) = \mathbf{A}\tilde{x}$ and $\tilde{v}_i = \tilde{z}_i$ gives the identity. □

Example 0.9 (Applying a matrix to a combination). Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\tilde{z}_1 = (1, 0)$ and $\tilde{z}_2 = (0, 1)$. Then $\mathbf{A}\tilde{z}_1 = (1, 3)$ and $\mathbf{A}\tilde{z}_2 = (2, 4)$, and for the combination $2\tilde{z}_1 - \tilde{z}_2 = (2, -1)$:

$$\begin{aligned} \mathbf{A}(2, -1) &= (1 \cdot 2 + 2 \cdot (-1), 3 \cdot 2 + 4 \cdot (-1)) && \text{(Definition 11 in Matrices)} \\ &= (0, 2) && \text{(arithmetic)} \\ &= 2(1, 3) - (2, 4), && \text{(arithmetic)} \end{aligned}$$

which is $2\mathbf{A}\tilde{z}_1 - \mathbf{A}\tilde{z}_2$, as Theorem 0.4 says.

Theorem 0.5 (More than p vectors in $\mathbb{R}^p$ are linearly dependent). *If $k > p$, then any k vectors $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$ are not linearly independent (Definition 0.1): some numbers $c_1, \dots, c_k$, not all zero, satisfy $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k = \tilde{0}_p$.*

Proof

Proof. **Induction** on p .

Base case, $p = 1$. Each $\tilde{v}_i$ is a single number v_i , and $k \geq 2$. If $v_1 = 0$, take $c_1 = 1$ and every other $c_i = 0$. Otherwise take $c_1 = v_2$, $c_2 = -v_1$ and every other $c_i = 0$; then $c_2 \neq 0$, and $v_2v_1 - v_1v_2 = 0$.

Inductive step. Let $p \geq 2$, assume the **theorem** holds for vectors in $\mathbb{R}^{p-1}$, and take $k > p$ vectors in $\mathbb{R}^p$. Write $v_{i,p}$ for the last entry of $\tilde{v}_i$.

Case 1: every $v_{i,p} = 0$. Deleting the last entry gives $k > p - 1$ vectors in $\mathbb{R}^{p-1}$, which by the inductive hypothesis have a combination with coefficients not all zero that equals $\tilde{0}_{p-1}$. The same coefficients combine the original vectors to $\tilde{0}_p$, since every last entry is 0.

Case 2: some $v_{i,p} \neq 0$. Renumber the vectors so that $v_{k,p} \neq 0$, and for $i = 1, \dots, k - 1$ let

$$\tilde{w}_i \stackrel{\text{def}}{=} \tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k.$$

The last entry of $\tilde{w}_i$ is $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$. Deleting that zero last entry gives $k - 1 > p - 1$ vectors in $\mathbb{R}^{p-1}$, so by the inductive hypothesis there are numbers $c_1, \dots, c_{k-1}$, not all zero, whose combination of the shortened vectors is $\tilde{0}_{p-1}$. Putting the zero last entries back gives $c_1\tilde{w}_1 + \dots + c_{k-1}\tilde{w}_{k-1} = \tilde{0}_p$. Then

$$\begin{aligned}
\tilde{0}_p &= c_1 \tilde{w}_1 + \cdots + c_{k-1} \tilde{w}_{k-1} && \text{(choice of } c_1, \dots, c_{k-1} \text{)} \\
&= \sum_{i=1}^{k-1} c_i \left(\tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k \right) && \text{(definition of } \tilde{w}_i \text{)} \\
&= \sum_{i=1}^{k-1} c_i \tilde{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \tilde{v}_k && \text{(distribute each } c_i, \text{ and split the sum)} \\
&= c_1 \tilde{v}_1 + \cdots + c_{k-1} \tilde{v}_{k-1} - \left(\sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \tilde{v}_k, && \text{(factor } \tilde{v}_k \text{ out of the second sum)}
\end{aligned}$$

a combination of $\tilde{v}_1, \dots, \tilde{v}_k$ equal to $\tilde{0}_p$ whose first $k-1$ coefficients are not all zero. $\square$

Example 0.10 (Three vectors in $\mathbb{R}^2$). Theorem 0.5 says the three vectors $(1, 0)$, $(0, 1)$ and $(1, 1)$ in $\mathbb{R}^2$ are linearly dependent, and indeed $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = (0, 0)$. The theorem says nothing about $k \leq p$ vectors: $(1, 0)$ and $(0, 1)$ are linearly independent, while $(1, 2)$ and $(2, 4)$ are not (Example 0.1).

Definition 0.6 (Basis). A **basis** of a subspace $\mathcal{S}$ (Definition 0.4) is a list of vectors that are linearly independent (Definition 0.1) and span $\mathcal{S}$ (Definition 0.5).

Example 0.11 (Two bases of $\mathbb{R}^2$).

- $(1, 0)$ and $(0, 1)$ form a basis of $\mathbb{R}^2$: they are linearly independent, and any $(x, y) \in \mathbb{R}^2$ equals $x(1, 0) + y(0, 1)$.
- $(1, 0)$ and $(1, 1)$ also form a basis of $\mathbb{R}^2$: Example 0.1 shows they are linearly independent, and any (x, y) equals $(x - y)(1, 0) + y(1, 1)$, because $(x - y) + y = x$ in the first entry and $0 + y = y$ in the second.

Example 0.12 (Lists that are not bases of $\mathbb{R}^2$).

- $(1, 0)$ and $(2, 0)$ are not a basis of $\mathbb{R}^2$. They are not linearly independent, since $2(1, 0) - (2, 0) = \tilde{0}$, and they do not span $\mathbb{R}^2$: every combination $c_1(1, 0) + c_2(2, 0)$ has second entry 0, so $(0, 1)$ is not in their span.
- $(1, 0)$, $(0, 1)$ and $(1, 1)$ span $\mathbb{R}^2$, but they are not a basis: $(1, 0) + (0, 1) - (1, 1) = \tilde{0}$, so they are not linearly independent.

Theorem 0.6 (Coefficients in a basis are unique). *If $\tilde{a}_1, \dots, \tilde{a}_k$ is a basis of a subspace $\mathcal{S}$ (Definition 0.6), then every $\tilde{x} \in \mathcal{S}$ is a linear combination $\tilde{x} = c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k$ for exactly one list of numbers $c_1, \dots, c_k$.*

i Proof

Proof. At least one such list exists, because the basis spans $\mathcal{S}$ (Definition 0.5). Suppose two lists both work:

$$\begin{aligned} c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k &= \tilde{x} \\ &= d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k. \end{aligned}$$

Subtracting the right side from the left side gives

$$\begin{aligned} \tilde{0} &= (c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k) - (d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k) \quad (\text{both combinations equal } \tilde{x}) \\ &= (c_1 - d_1)\tilde{a}_1 + \dots + (c_k - d_k)\tilde{a}_k \quad (\text{regroup, and distribute each } \tilde{a}_i) \end{aligned}$$

Because $\tilde{a}_1, \dots, \tilde{a}_k$ are linearly independent (Definition 0.1), every coefficient $c_i - d_i$ is 0, so $c_i = d_i$ for each i . $\square$

Example 0.13 (Coefficients of $(3, 5)$ in two bases). In the basis $(1, 0), (0, 1)$ of Example 0.11, $(3, 5) = 3(1, 0) + 5(0, 1)$, and no other coefficients work. In the basis $(1, 0), (1, 1)$, the formula in Example 0.11 gives

$$\begin{aligned} (3, 5) &= (3 - 5)(1, 0) + 5(1, 1) \\ &= -2(1, 0) + 5(1, 1). \end{aligned}$$

The same vector has different coefficients in different bases, but within one basis its coefficients are unique.

The list $(1, 0), (0, 1), (1, 1)$ in Example 0.12 spans $\mathbb{R}^2$ but is not linearly independent, and the coefficients of $(3, 5)$ are not unique:

$$\begin{aligned} (3, 5) &= 3(1, 0) + 5(0, 1) + 0(1, 1) \\ &= 0(1, 0) + 2(0, 1) + 3(1, 1). \end{aligned}$$

Definition 0.7 (Coordinates in a basis). Let $\tilde{a}_1, \dots, \tilde{a}_k$ be a basis of a subspace $\mathcal{S}$ (Definition 0.6), and let $\tilde{x} \in \mathcal{S}$. The **coordinates** of $\tilde{x}$ in this basis are the numbers $c_1, \dots, c_k$ with $\tilde{x} = c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k$, and $(c_1, \dots, c_k) \in \mathbb{R}^k$ is the **coordinate vector** of $\tilde{x}$. By Theorem 0.6 there is exactly one such list of numbers.

Example 0.14 (Coordinates of $(4, 1)$ in two bases of $\mathbb{R}^2$). In the basis $(1, 0), (1, 1)$ of Example 0.11, the coordinates of $(4, 1)$ are 3 and 1:

$$\begin{aligned} 3(1, 0) + 1(1, 1) &= (3 + 1, 0 + 1) \quad (\text{Definition 8 in Vectors, entry by entry}) \\ &= (4, 1). \quad (\text{add}) \end{aligned}$$

In the basis $(1, 0), (0, 1)$, its coordinates are its entries, 4 and 1.

Theorem 0.7 (All bases of a subspace have the same number of vectors). *If $\tilde{a}_1, \dots, \tilde{a}_k$ and $\tilde{b}_1, \dots, \tilde{b}_l$ are both bases of a subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 0.6), then $k = l$.*

i Proof

Proof. Let $\mathbf{A}$ be the $p \times k$ matrix with columns $\tilde{a}_1, \dots, \tilde{a}_k$, and $\mathbf{B}$ the $p \times l$ matrix with columns $\tilde{b}_1, \dots, \tilde{b}_l$.

Write $\mathbf{A}$ in terms of $\mathbf{B}$. Each $\tilde{a}_j$ is in $\mathcal{S}$, and the $\tilde{b}$'s span $\mathcal{S}$ (Definition 0.5), so $\tilde{a}_j = \mathbf{B}\tilde{c}_j$ for some $\tilde{c}_j \in \mathbb{R}^l$ (Definition 11 in Matrices). Let $\mathbf{C}$ be the $l \times k$ matrix with columns $\tilde{c}_1, \dots, \tilde{c}_k$. Then $\underset{p \times k}{\mathbf{A}} = \underset{p \times l}{\mathbf{B}} \underset{l \times k}{\mathbf{C}}$, because column j of $\mathbf{BC}$ is $\mathbf{B}\tilde{c}_j$.

The columns of $\mathbf{C}$ are linearly independent. Suppose $\mathbf{C}\tilde{x} = \tilde{0}$ for some $\tilde{x} \in \mathbb{R}^k$. Then

$$\begin{aligned} \mathbf{A}\tilde{x} &= (\mathbf{BC})\tilde{x} \quad (\text{substitute } \mathbf{A} = \mathbf{BC}) \\ &= \mathbf{B}(\mathbf{C}\tilde{x}) \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{B}\tilde{0} \quad (\text{substitute } \mathbf{C}\tilde{x} = \tilde{0}) \\ &= \tilde{0} \quad (\text{a matrix times the zero vector is } \tilde{0}) \end{aligned}$$

$\mathbf{A}\tilde{x}$ is the combination $x_1\tilde{a}_1 + \dots + x_k\tilde{a}_k$ (Theorem 0.3), and the $\tilde{a}$'s are linearly independent (Definition 0.1), so $\tilde{x} = \tilde{0}$. So the only combination of the columns of $\mathbf{C}$ that equals $\tilde{0}$ has all coefficients 0.

Count. $\mathbf{C}$ has k linearly independent columns of length l . More than l vectors in $\mathbb{R}^l$ are never linearly independent (Theorem 0.5), so $k \leq l$. Exchanging the roles of the two bases gives $l \leq k$, so $k = l$. $\square$

Example 0.15 (Both bases of $\mathbb{R}^2$ in Example 0.11 have two vectors). The bases $(1, 0), (0, 1)$ and $(1, 0), (1, 1)$ of $\mathbb{R}^2$ in Example 0.11 both have 2 vectors, as Theorem 0.7 requires. So no list of 3 vectors is a basis of $\mathbb{R}^2$, which agrees with Example 0.12: the list $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is not linearly independent.

Definition 0.8 (Dimension). The **dimension** of a subspace $\mathcal{S}$, written $\dim(\mathcal{S})$, is the number of vectors in any basis of $\mathcal{S}$ (Definition 0.6). Theorem 0.7 makes this number well defined. By convention, the empty list is the basis of $\{\tilde{0}\}$, so $\dim(\{\tilde{0}\}) = 0$.

Example 0.16 (The dimensions of some subspaces).

- $\dim(\{\tilde{0}\}) = 0$, by the convention in Definition 0.8.
- $\dim(\mathbb{R}^2) = 2$, by Example 0.11. More generally, $\dim(\mathbb{R}^p) = p$: the indicator vectors $\tilde{e}_1, \dots, \tilde{e}_p$ (Definition 12 in Vectors) are linearly independent, since $c_1\tilde{e}_1 + \dots + c_p\tilde{e}_p = (c_1, \dots, c_p)$, which is $\tilde{0}_p$ only if every $c_i = 0$; and they span $\mathbb{R}^p$, since any $\tilde{x} = x_1\tilde{e}_1 + \dots + x_p\tilde{e}_p$.
- The line in Example 0.3 has dimension 1: the single vector $(1, 2)$ is linearly independent and spans the line.
- The plane $z = 0$ in Example 0.6 has dimension 2, with basis $(1, 0, 0), (0, 1, 0)$.

Theorem 0.8 (Adding a vector outside the span keeps a list independent). *If $\tilde{v}_1, \dots, \tilde{v}_k$ are linearly independent (Definition 0.1) and $\tilde{w}$ is not in $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$ (Definition 0.5), then $\tilde{v}_1, \dots, \tilde{v}_k, \tilde{w}$ are linearly independent.*

i Proof

Proof. Suppose $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k + d\tilde{w} = \tilde{0}$.

The coefficient d is 0. If $d \neq 0$, then

$$\begin{aligned} \tilde{w} &= \frac{1}{d}(d\tilde{w}) && (d \neq 0) \\ &= \frac{1}{d}(-c_1\tilde{v}_1 - \dots - c_k\tilde{v}_k) && (\text{solve the supposed equation for } d\tilde{w}) \\ &= \left(-\frac{c_1}{d}\right)\tilde{v}_1 + \dots + \left(-\frac{c_k}{d}\right)\tilde{v}_k, && (\text{distribute } \frac{1}{d}) \end{aligned}$$

which puts $\tilde{w}$ in $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$, contrary to the assumption.

The other coefficients are 0. With $d = 0$, the equation reads $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k = \tilde{0}$, so every $c_i = 0$, because $\tilde{v}_1, \dots, \tilde{v}_k$ are linearly independent. $\square$

Example 0.17 (Adding $(0, 0, 1)$ to two independent vectors in $\mathbb{R}^3$). The vectors $(1, 0, 0)$ and $(0, 1, 0)$ are linearly independent, and their span is the plane $z = 0$ (Example 0.6). The vector $(0, 0, 1)$ has third entry 1, so it is not in that plane, and by Theorem 0.8 the three vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ are linearly independent. Adding $(1, 1, 0)$ instead would not work: $(1, 1, 0)$ is in the span, and $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$.

Theorem 0.9 (An independent list extends to a basis). *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.4) and $\tilde{v}_1, \dots, \tilde{v}_k \in \mathcal{S}$ are linearly independent (Definition 0.1), then adding vectors of $\mathcal{S}$ to the list gives a basis of $\mathcal{S}$ (Definition 0.6). Starting from the empty list shows that every subspace has a basis.*

i Proof

Proof. Repeat the following step. If the current list spans $\mathcal{S}$, stop: it is linearly independent and spans $\mathcal{S}$, so it is a basis. Otherwise, some $\tilde{w} \in \mathcal{S}$ is not in the span of the current list; add $\tilde{w}$ to the list, which stays linearly independent by Theorem 0.8.

Every list produced is a linearly independent list of vectors in $\mathbb{R}^p$, so it has at most p vectors (Theorem 0.5). Each step adds one vector, so the process stops after at most $p - k$ steps, and when it stops the list is a basis of $\mathcal{S}$.

For the empty list, which is linearly independent and spans $\{\tilde{0}\}$ by the conventions in Definition 0.5, the first step asks whether $\mathcal{S} = \{\tilde{0}\}$: if so, the empty list is already a basis; if not, the process adds vectors as described. $\square$

Example 0.18 (Extending $(1, 1, 0)$ to a basis of $\mathbb{R}^3$). Start with the single vector $(1, 1, 0)$.

1. $(1, 0, 0)$ is not in $\text{span}\{(1, 1, 0)\}$, because every multiple $c(1, 1, 0)$ has equal first and second entries. Add it.
2. $(0, 0, 1)$ is not in $\text{span}\{(1, 1, 0), (1, 0, 0)\}$, because every combination of those two vectors has third entry 0. Add it.
3. The list $(1, 1, 0), (1, 0, 0), (0, 0, 1)$ spans $\mathbb{R}^3$: $(x, y, z) = y(1, 1, 0) + (x - y)(1, 0, 0) + z(0, 0, 1)$. Stop.

The result is a basis of $\mathbb{R}^3$ that contains the starting vector.

Theorem 0.10 (An independent list in a subspace has at most $\dim$ vectors). *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = d$ (Definition 0.8).*

1. *Any linearly independent list of vectors in $\mathcal{S}$ has at most d vectors.*
2. *If $\mathcal{T}$ is a subspace with $\mathcal{T} \subseteq \mathcal{S}$, then $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$.*

i Proof

Proof. Part 1. If $d = 0$, then $\mathcal{S} = \{\tilde{0}\}$, so every vector in a nonempty list from $\mathcal{S}$ is $\tilde{0}$, and $1 \cdot \tilde{0} = \tilde{0}$ shows the list is not linearly independent; only the empty list, with $0 = d$ vectors, is. Now let $d \geq 1$, and let $\tilde{a}_1, \dots, \tilde{a}_d$ be a basis of $\mathcal{S}$, and let $\tilde{u}_1, \dots, \tilde{u}_k \in \mathcal{S}$ be

linearly independent. Each $\tilde{u}_j$ is in $\mathcal{S}$, so $\tilde{u}_j = c_{1j}\tilde{a}_1 + \cdots + c_{dj}\tilde{a}_d$ for some coordinate vector $\tilde{c}_j = (c_{1j}, \dots, c_{dj}) \in \mathbb{R}^d$. Suppose $k > d$. Then $\tilde{c}_1, \dots, \tilde{c}_k$ are $k > d$ vectors in $\mathbb{R}^d$, so some $x_1, \dots, x_k$, not all zero, satisfy $x_1\tilde{c}_1 + \cdots + x_k\tilde{c}_k = \tilde{0}_d$ (Theorem 0.5); entry i of that equation says $\sum_{j=1}^k x_j c_{ij} = 0$. Then

$$\begin{aligned}
\sum_{j=1}^k x_j \tilde{u}_j &= \sum_{j=1}^k x_j \sum_{i=1}^d c_{ij} \tilde{a}_i && \text{(substitute each } \tilde{u}_j \text{)} \\
&= \sum_{j=1}^k \sum_{i=1}^d x_j c_{ij} \tilde{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^d \sum_{j=1}^k x_j c_{ij} \tilde{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^d \left(\sum_{j=1}^k x_j c_{ij} \right) \tilde{a}_i && \text{(factor } \tilde{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^d 0 \cdot \tilde{a}_i && \text{(entry } i \text{ of } \sum_j x_j \tilde{c}_j = \tilde{0}_d \text{)} \\
&= \tilde{0}_p, && \text{(a zero multiple of a vector is } \tilde{0} \text{)}
\end{aligned}$$

a combination of $\tilde{u}_1, \dots, \tilde{u}_k$ with coefficients not all zero that equals $\tilde{0}_p$. That contradicts their linear independence, so $k \leq d$.

Part 2. $\mathcal{T}$ has a basis (Theorem 0.9) with $\dim(\mathcal{T})$ vectors. Those vectors are linearly independent and lie in $\mathcal{S}$, so by Part 1 there are at most $\dim(\mathcal{S})$ of them. $\square$

Example 0.19 (Subspaces of a plane have dimension at most two). The plane $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ in $\mathbb{R}^3$ has dimension 2 (Example 0.16). By Theorem 0.10, no three vectors with third entry 0 are linearly independent; for instance, $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$. The line $\mathcal{T} = \text{span}\{(1, 1, 0)\}$ lies in $\mathcal{S}$, and

$$\begin{aligned}
\dim(\mathcal{T}) &= 1 \\
&\leq 2 \\
&= \dim(\mathcal{S}).
\end{aligned}$$

Theorem 0.11 (A subspace inside another of the same dimension equals it). *If $\mathcal{T}$ and $\mathcal{S}$ are subspaces of $\mathbb{R}^p$ with $\mathcal{T} \subseteq \mathcal{S}$ and $\dim(\mathcal{T}) = \dim(\mathcal{S})$ (Definition 0.8), then $\mathcal{T} = \mathcal{S}$.*

Proof

Proof. Let $d = \dim(\mathcal{S})$, and let $\tilde{t}_1, \dots, \tilde{t}_d$ be a basis of $\mathcal{T}$ (Theorem 0.9); it has d vectors because $\dim(\mathcal{T}) = d$. (When $d = 0$ it is the empty list, whose span is $\{\tilde{0}\}$ by the convention in Definition 0.5, and the argument is unchanged.) Suppose some $\tilde{s} \in \mathcal{S}$ is not in $\text{span}\{\tilde{t}_1, \dots, \tilde{t}_d\} = \mathcal{T}$. Then $\tilde{t}_1, \dots, \tilde{t}_d, \tilde{s}$ are linearly independent (Theorem 0.8), and they are $d + 1$ vectors of $\mathcal{S}$, which contradicts part 1 of Theorem 0.10. So every vector of $\mathcal{S}$ is in $\mathcal{T}$, that is, $\mathcal{S} \subseteq \mathcal{T}$. Together with $\mathcal{T} \subseteq \mathcal{S}$, this inclusion gives $\mathcal{T} = \mathcal{S}$. $\square$

Example 0.20 (Two independent vectors in a plane span it). Let $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$, the plane $z = 0$, which has dimension 2 (Example 0.16), and let $\mathcal{T} = \text{span}\{(1, 0, 0), (1, 1, 0)\}$. Both vectors have third entry 0, so $\mathcal{T} \subseteq \mathcal{S}$. They are linearly independent: $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$, which is $\tilde{0}$ only if $c_2 = 0$ and then $c_1 = 0$. So $\dim(\mathcal{T}) = 2$, and Theorem 0.11 gives $\mathcal{T} = \mathcal{S}$ without solving for any coefficients. As a check, $(x, y, 0) = (x - y)(1, 0, 0) + y(1, 1, 0)$.

Example 0.21 (Equal dimensions without containment). The lines $\text{span}\{(1, 0, 0)\}$ and $\text{span}\{(0, 1, 0)\}$ in $\mathbb{R}^3$ both have dimension 1, but they are not equal: $(1, 0, 0)$ is in the first and not the second. Theorem 0.11 does not apply, because neither line contains the other.

Column space and null space

Definition 0.9 (Column space (range, image)). The **column space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\tilde{x} : \tilde{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\tilde{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$. The **row space** of $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.22 (The column space of a rank-one matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\tilde{x} \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{A}\tilde{x} &= (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)(1, 3) \end{aligned}$$

([Definition 11 in Matrices](#)). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.12 (The column space is the span of the columns). *If $\mathbf{A}$ is an $m \times n$ matrix with columns $\tilde{a}_1, \dots, \tilde{a}_n$, then*

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\tilde{a}_1, \dots, \tilde{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$, and the row space $\mathcal{C}(\mathbf{A}^\top)$ is a subspace of $\mathbb{R}^n$.

i Proof

Proof. By Theorem [0.3](#), $\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n$, and as $\tilde{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \dots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\tilde{x}$ (Definition [0.9](#)) is the set of all linear combinations of the columns, which is their span (Definition [0.5](#)). A span is a subspace (Theorem [0.2](#)). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. $\square$

Example 0.23 (Reading off the column space and row space from the columns and rows). For $\mathbf{A}$ in Example [0.22](#), the columns are $(1, 3)$, $(-2, -6) = -2(1, 3)$ and $(-2, -6) = -2(1, 3)$, so by Theorem [0.12](#) $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$, the same line found in Example [0.22](#). The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3(1, -2, -2)$, so the row space is $\text{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.10 (Null space (kernel)). The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^n : \mathbf{A}\tilde{x} = \tilde{0}_m\},$$

the set of [solutions](#) of the linear system $\mathbf{A}\tilde{x} = \tilde{0}_m$. It is also called the **kernel** of $\mathbf{A}$. The **left null space** of $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.24 (The null space and left null space of a rank-one matrix). For $\mathbf{A}$ in Example [0.22](#), $\mathbf{A}\tilde{x} = (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\tilde{x} = \tilde{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned} \tilde{x} &= (2x_2 + 2x_3, x_2, x_3) \\ &= x_2(2, 1, 0) + x_3(2, 0, 1) \end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$, a plane in $\mathbb{R}^3$.

For the left null space, $\mathbf{A}^\top \tilde{\mathbf{y}} = (y_1 + 3y_2, -2(y_1 + 3y_2), -2(y_1 + 3y_2))$, which is $\tilde{\mathbf{0}}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$, a line in $\mathbb{R}^2$.

Theorem 0.13 (A null space is a subspace). *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ (Definition 0.10) is a subspace of $\mathbb{R}^n$ (Definition 0.4).*

i Proof

Proof. $\mathbf{A}\tilde{\mathbf{0}}_n = \tilde{\mathbf{0}}_m$, so $\tilde{\mathbf{0}}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\tilde{\mathbf{u}}, \tilde{\mathbf{w}} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned} \mathbf{A}(\tilde{\mathbf{u}} + \tilde{\mathbf{w}}) &= \mathbf{A}\tilde{\mathbf{u}} + \mathbf{A}\tilde{\mathbf{w}} && \text{(Theorem 6 in Matrices)} \\ &= \tilde{\mathbf{0}}_m + \tilde{\mathbf{0}}_m && (\tilde{\mathbf{u}} \text{ and } \tilde{\mathbf{w}} \text{ are in the null space)} \\ &= \tilde{\mathbf{0}}_m && (\text{adding } \tilde{\mathbf{0}}_m \text{ changes nothing}) \end{aligned}$$

and, for each entry $i = 1, \dots, m$,

$$\begin{aligned} (\mathbf{A}(c\tilde{\mathbf{u}}))_i &= a_{i1}(cu_1) + \dots + a_{in}(cu_n) && \text{(Definition 11 in Matrices)} \\ &= c(a_{i1}u_1 + \dots + a_{in}u_n) && (\text{factor } c \text{ out of each term}) \\ &= c(\mathbf{A}\tilde{\mathbf{u}})_i && \text{(Definition 11 in Matrices)} \\ &= c \cdot 0 && (\tilde{\mathbf{u}} \text{ is in the null space)} \\ &= 0, && (\text{arithmetic}) \end{aligned}$$

so $\mathbf{A}(c\tilde{\mathbf{u}}) = \tilde{\mathbf{0}}_m$. Therefore $\tilde{\mathbf{u}} + \tilde{\mathbf{w}}$ and $c\tilde{\mathbf{u}}$ are in $\mathcal{N}(\mathbf{A})$. □

Example 0.25 (The solutions of $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ with $\tilde{\mathbf{b}} \neq \tilde{\mathbf{0}}$ are not a subspace). In Example 0.24, $(2, 1, 0)$ and $(2, 0, 1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4, 1, 1)$, since $4 - 2 \cdot 1 - 2 \cdot 1 = 0$. By contrast, the solutions of $\mathbf{A}\tilde{\mathbf{x}} = (1, 3)$ for the same $\mathbf{A}$ are the vectors with $x_1 - 2x_2 - 2x_3 = 1$. That set does not contain $\tilde{\mathbf{0}}_3$, so by Theorem 0.1 it is not a subspace.

Theorem 0.14 (The row space and null space share only the zero vector). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\tilde{\mathbf{0}}_n\}.$$

i Proof

Proof. Both sets are subspaces, by Theorem 0.12 and Theorem 0.13, so both contain $\tilde{\mathbf{0}}_n$ (Theorem 0.1).

Now take any $\tilde{x}$ in both sets. Being in the row space means $\tilde{x} = \mathbf{A}^\top \tilde{u}$ for some $\tilde{u} \in \mathbb{R}^m$ (Definition 0.9), and being in the null space means $\mathbf{A}\tilde{x} = \tilde{\mathbf{0}}_m$ (Definition 0.10). Then

$$\begin{aligned}\tilde{x}^\top \tilde{x} &= (\mathbf{A}^\top \tilde{u})^\top \tilde{x} && \text{(substitute } \tilde{x} = \mathbf{A}^\top \tilde{u} \text{ in the first factor)} \\ &= (\tilde{u}^\top (\mathbf{A}^\top)^\top) \tilde{x} && \text{(Theorem 16 in Matrices)} \\ &= (\tilde{u}^\top \mathbf{A}) \tilde{x} && \text{(transposing twice returns the original matrix)} \\ &= \tilde{u}^\top (\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\ &= \tilde{u}^\top \tilde{\mathbf{0}}_m && \text{(substitute } \mathbf{A}\tilde{x} = \tilde{\mathbf{0}}_m \text{)} \\ &= 0. && \text{(every term of the inner product is 0)}\end{aligned}$$

But $\tilde{x}^\top \tilde{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\tilde{x} = \tilde{\mathbf{0}}_n$. □

Example 0.26 (The row space and null space in Example 0.24). For $\mathbf{A}$ in Example 0.22, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.24). A vector $c(1, -2, -2)$ of the row space is in the null space only if

$$\begin{aligned}\mathbf{A}(c(1, -2, -2)) &= c(1 - 2 \cdot (-2) - 2 \cdot (-2), 3(1 - 2 \cdot (-2) - 2 \cdot (-2))) && \text{(Example 0.24's formula for } \mathbf{A}\tilde{x} \text{)} \\ &= c(9, 27) && \text{(arithmetic)}\end{aligned}$$

is $\tilde{\mathbf{0}}_2$, that is, only if $c = 0$. So the two subspaces share only $\tilde{\mathbf{0}}_3$.

Definition 0.11 (Gram matrix). The **Gram matrix** of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$.

Example 0.27 (A Gram matrix and its entries). Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\tilde{a}_1 = (1, 1, 0)$ and $\tilde{a}_2 = (0, 1, 1)$. Then

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && \text{(Definition 3 in Matrices)} \\
&= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} && \text{(Definition 7 in Matrices)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i, j) of the Gram matrix is the dot product $\tilde{a}_i \cdot \tilde{a}_j$: for instance,

$$\begin{aligned}
\tilde{a}_1 \cdot \tilde{a}_2 &= 0 + 1 + 0 \\
&= 1
\end{aligned}$$

is entry $(1, 2)$.

Theorem 0.15 ($\mathbf{A}^\top \mathbf{A}$ has the same null space as $\mathbf{A}$). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

i Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$. If $\mathbf{A}\tilde{x} = \tilde{0}_m$, then

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A}\tilde{x} &= \mathbf{A}^\top \tilde{0}_m \\
&= \tilde{0}_n.
\end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$. If $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_n$, then

$$\begin{aligned}
0 &= \tilde{x}^\top \tilde{0}_n && \text{(every term of the inner product is 0)} \\
&= \tilde{x}^\top (\mathbf{A}^\top \mathbf{A}\tilde{x}) && \text{(substitute } \tilde{0}_n = \mathbf{A}^\top \mathbf{A}\tilde{x}) \\
&= (\tilde{x}^\top \mathbf{A}^\top) (\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{A}\tilde{x})^\top (\mathbf{A}\tilde{x}) && \text{(Theorem 16 in Matrices)} \\
&= (\mathbf{A}\tilde{x}) \cdot (\mathbf{A}\tilde{x}) && \text{(Example 6 in Matrices)} \\
&= \|\mathbf{A}\tilde{x}\|^2. && \text{(square both sides of Equation 2 in Vectors)}
\end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\tilde{x} = \tilde{0}_m$. □

Example 0.28 (The null space of a Gram matrix). For $\mathbf{A}$ in Example 0.22,

$$\begin{aligned}\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.\end{aligned}$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top \mathbf{A} \tilde{x} = \tilde{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.24.

Rank-nullity and rank factorization

i Source

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Theorem 0.16 (The rank is the dimension of the column space). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

i Proof

Proof. Let $r = \text{rank}(\mathbf{A})$ (Definition 0.2), and choose r linearly independent columns of $\mathbf{A}$; by Definition 0.2, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 0.6).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\tilde{a}_j$ were not in the span, then adding it to the chosen columns would give $r + 1$ linearly independent columns (Theorem 0.8), more than r . So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 0.2), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.12, those combinations make up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.

The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 0.8). If $r = 0$, every column is $\tilde{0}_m$, $\mathcal{C}(\mathbf{A}) = \{\tilde{0}_m\}$, and both sides are 0. $\square$

Example 0.29 (The rank of the matrix in Example 0.22). For $\mathbf{A}$ in Example 0.22, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (1), and $(1, 3)$ is linearly independent because it is nonzero, so it is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.16. Directly: the first column $(1, 3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1, 3)$.

Definition 0.12 (Nullity). The **nullity** of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.10):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.30 (The nullity of the matrix in Example 0.22). For $\mathbf{A}$ in Example 0.22, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.24). Those two vectors are linearly independent: $c_1(2, 1, 0) + c_2(2, 0, 1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\tilde{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\tilde{x} = \tilde{x}$ is $\tilde{0}_2$ only when $\tilde{x} = \tilde{0}_2$.

Theorem 0.17 (Rank-nullity theorem). For any $m \times n$ matrix $\mathbf{A}$,

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

i Proof

Proof. Let $\nu = \text{nullity}(\mathbf{A})$. Choose a basis $\tilde{x}_1, \dots, \tilde{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\tilde{x}_1, \dots, \tilde{x}_\nu, \tilde{y}_1, \dots, \tilde{y}_{n-\nu}$ of $\mathbb{R}^n$ (Theorem 0.9); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors (Theorem 0.7, Example 0.16). We show that $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using Theorem 0.4 to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \mathbf{A}\tilde{y}_i = \tilde{0}_m$. Then $\mathbf{A}(\sum_i v_i \tilde{y}_i) = \tilde{0}_m$ (Theorem 0.4), so $\sum_i v_i \tilde{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^\nu u_j \tilde{x}_j$ for some numbers u_j , since the $\tilde{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^{\nu} (-u_j) \tilde{x}_j + \sum_{i=1}^{n-\nu} v_i \tilde{y}_i = \tilde{0}_n,$$

and because the $\tilde{x}$'s and $\tilde{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\tilde{w} \in \mathcal{C}(\mathbf{A})$, so $\tilde{w} = \mathbf{A}\tilde{z}$ for some $\tilde{z} \in \mathbb{R}^n$ (Definition 0.9). Write $\tilde{z} = \sum_j a_j \tilde{x}_j + \sum_i b_i \tilde{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned} \tilde{w} &= \mathbf{A} \left(\sum_{j=1}^{\nu} a_j \tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \tilde{y}_i \right) \quad (\text{substitute } \tilde{z}) \\ &= \sum_{j=1}^{\nu} a_j \mathbf{A}\tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i \quad (\text{Theorem 0.4}) \\ &= \sum_{j=1}^{\nu} a_j \tilde{0}_m + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i \quad (\text{each } \tilde{x}_j \in \mathcal{N}(\mathbf{A})) \\ &= \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i, \quad (\text{drop the zero terms}) \end{aligned}$$

a linear combination of $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.

So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\text{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.16. $\square$

Example 0.31 (Checking rank-nullity on the matrix in Example 0.22). The matrix $\mathbf{A}$ in Example 0.22 has $n = 3$ columns, rank 1 (Example 0.29) and nullity 2 (Example 0.30), and $1 + 2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.30), and $2 + 0 = 2$.

Theorem 0.18 (Multiplying on the right cannot increase the rank). *If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times k$, then*

$$\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A}).$$

i Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$. Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\tilde{x}$ for some $\tilde{x} \in \mathbb{R}^k$ (Definition 0.9), and

$$(\mathbf{AB})\tilde{x} = \mathbf{A}(\mathbf{B}\tilde{x}) \quad (\text{Theorem 5 in Matrices})$$

is $\mathbf{A}$ times the vector $\mathbf{B}\tilde{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.

Compare dimensions. Both column spaces are subspaces (Theorem 0.12), so

$$\begin{aligned}
\text{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) \quad (\text{Theorem 0.16}) \\
&\leq \dim(\mathcal{C}(\mathbf{A})) \quad (\text{Theorem 0.10, part 2}) \\
&= \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.16})
\end{aligned}$$

□

Example 0.32 (A product can lose rank). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{AB} = \begin{bmatrix} 1 \cdot 0 + 0 \cdot 1 & 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.18 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Theorem 0.19 (Fundamental theorem of ranks). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\begin{aligned}
\text{rank}(\mathbf{A}) &= \text{rank}(\mathbf{A}^\top) \\
&= \text{rank}(\mathbf{A}^\top \mathbf{A}) \\
&= \text{rank}(\mathbf{AA}^\top).
\end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ equals the largest number of linearly independent columns.

i Proof

Proof. $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$. Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.15), so they have the same nullity (Definition 0.12). Then

$$\begin{aligned}
\text{rank}(\mathbf{A}^\top \mathbf{A}) &= n - \text{nullity}(\mathbf{A}^\top \mathbf{A}) \quad (\text{Theorem 0.17}) \\
&= n - \text{nullity}(\mathbf{A}) \quad (\text{equal null spaces}) \\
&= \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.17})
\end{aligned}$$

$\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A}^\top)$. Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned}
\text{rank}(\mathbf{A}) &= \text{rank}(\mathbf{A}^\top \mathbf{A}) \quad (\text{first step}) \\
&\leq \text{rank}(\mathbf{A}^\top) \quad (\text{Theorem 0.18, with } \mathbf{A}^\top \text{ on the left}) \\
&= \text{rank}(\mathbf{AA}^\top) \quad (\text{second step}) \\
&\leq \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.18, with } \mathbf{A} \text{ on the left})
\end{aligned}$$

The chain starts and ends at $\text{rank}(\mathbf{A})$, so every quantity in it equals $\text{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. $\square$

Example 0.33 (Four equal ranks). For $\mathbf{A}$ in Example 0.22, $\text{rank}(\mathbf{A}) = 1$ (Example 0.29).

Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top \mathbf{A}$ in Example 0.28 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \begin{bmatrix} 1 \cdot 1 + (-2)(-2) + (-2)(-2) & 1 \cdot 3 + (-2)(-6) + (-2)(-6) \\ 3 \cdot 1 + (-6)(-2) + (-6)(-2) & 3 \cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix}, \end{aligned}$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension). *For any $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$.*

i Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\text{rank}(\mathbf{A}) \leq n$ (Definition 0.2). Likewise $\mathbf{A}^\top$ has m columns, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) \leq m$ (Theorem 0.19). $\square$

Example 0.34 (A 2×3 matrix has rank at most 2). By Corollary 0.1, every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Remark 0.1 reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Definition 0.13 (Affine subspace). An **affine subspace** of $\mathbb{R}^p$ is a set of the form

$$\{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{S}\}$$

for some vector $\tilde{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 0.4): the **image** of $\mathcal{S}$ under the translation by $\tilde{x}_0$ (Definition 16 in Matrices).

Example 0.35 (A line that misses the origin is an affine subspace). The line $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ of Example 0.4 is not a subspace, but it is an affine subspace, with $\tilde{x}_0 = (0, 1)$ and $\mathcal{S} = \text{span}\{(1, 0)\}$ (Definition 0.5):

$$\begin{aligned} (0, 1) + c(1, 0) &= (0 + c, 1 + 0) \quad (\text{Definition 6 in Matrices, Definition 6 in Vectors}) \\ &= (c, 1), \quad (\text{add}) \end{aligned}$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\tilde{x}_0 = \tilde{0}$.

Theorem 0.20 (A hyperplane is a translated subspace of dimension $p - 1$). *Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} + b = 0\}$ be the hyperplane with normal vector $\tilde{w}$ and offset b (Definition 19 in Matrices), and let $\mathcal{H}_0 = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ (Definition 0.4) of dimension $p - 1$ (Definition 0.8).
2. $\mathcal{H}$ is not empty, and for any $\tilde{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.13).
3. For any $\tilde{x}, \tilde{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\tilde{w} \perp (\tilde{x} - \tilde{y})$ (Definition 13 in Vectors).

i Proof

Proof. Here $\tilde{w}^\top$ is a $1 \times p$ matrix, and $\tilde{w}^\top \tilde{x} = \tilde{w} \cdot \tilde{x}$ (Example 6 in Matrices).

Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\tilde{w}^\top)$ (Definition 0.10), so it is a subspace (Theorem 0.13). The p columns of $\tilde{w}^\top$ are the numbers $w_1, \dots, w_p$. Some $w_j \neq 0$, because $\tilde{w} \neq \tilde{0}$, and that column on its own is linearly independent ($c w_j = 0$ forces $c = 0$), so $\text{rank}(\tilde{w}^\top) \geq 1$ (Definition 0.2); and $\text{rank}(\tilde{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$\begin{aligned} \dim(\mathcal{H}_0) &= \text{nullity}(\tilde{w}^\top) \quad (\text{Definition 0.12}) \\ &= p - \text{rank}(\tilde{w}^\top) \quad (\text{Theorem 0.17, for the } p \text{ columns of } \tilde{w}^\top) \\ &= p - 1. \quad (\text{the rank is 1}) \end{aligned}$$

Part 2. $\tilde{w} \cdot \tilde{w} = w_1^2 + \dots + w_p^2 > 0$, because some $w_j \neq 0$. So $\tilde{x}^* \stackrel{\text{def}}{=} -\frac{b}{\tilde{w} \cdot \tilde{w}} \tilde{w}$ is defined, and

$$\begin{aligned} \tilde{w}^\top \tilde{x}^* + b &= -\frac{b}{\tilde{w} \cdot \tilde{w}} (\tilde{w}^\top \tilde{w}) + b \quad (\text{homogeneity of } \tilde{x} \mapsto \tilde{w}^\top \tilde{x}, \text{ Theorem 9 in Matrices}) \\ &= -b + b \quad (\tilde{w}^\top \tilde{w} = \tilde{w} \cdot \tilde{w}) \\ &= 0, \quad (\text{arithmetic}) \end{aligned}$$

so $\tilde{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\tilde{x}_0 \in \mathcal{H}$, so $\tilde{w}^\top \tilde{x}_0 = -b$. For any $\tilde{x} \in \mathbb{R}^p$,

$$\begin{aligned}\tilde{w}^\top (\tilde{x} - \tilde{x}_0) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{x}_0 \quad (\text{Theorem 0.4, with coefficients 1 and } -1) \\ &= \tilde{w}^\top \tilde{x} - (-b) \quad (\tilde{x}_0 \in \mathcal{H}) \\ &= \tilde{w}^\top \tilde{x} + b, \quad (\text{arithmetic})\end{aligned}$$

so $\tilde{x} \in \mathcal{H}$ exactly when $\tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$, that is, exactly when $\tilde{x} = \tilde{x}_0 + \tilde{s}$ with $\tilde{s} = \tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.13.

Part 3. For $\tilde{x}, \tilde{y} \in \mathcal{H}$, both $\tilde{w}^\top \tilde{x}$ and $\tilde{w}^\top \tilde{y}$ equal $-b$, so

$$\begin{aligned}\tilde{w} \cdot (\tilde{x} - \tilde{y}) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{y} \quad (\text{Theorem 0.4, with coefficients 1 and } -1) \\ &= (-b) - (-b) \quad (\tilde{x}, \tilde{y} \in \mathcal{H}) \\ &= 0. \quad (\text{arithmetic})\end{aligned}$$

□

Example 0.36 (The plane $x_1 + x_2 + x_3 = 1$ in $\mathbb{R}^3$). Take $\tilde{w} = (1, 1, 1)$ and $b = -1$, so $\mathcal{H} = \{\tilde{x} : x_1 + x_2 + x_3 - 1 = 0\}$, the plane that Example 0.5 showed is not a subspace. By part 1, $\mathcal{H}_0 = \{\tilde{x} : x_1 + x_2 + x_3 = 0\}$ is a subspace of dimension $3 - 1 = 2$.

- $\tilde{x}_0 = (1, 0, 0)$ is in $\mathcal{H}$, since $1 + 0 + 0 - 1 = 0$, and so is $\tilde{y} = (0, 1, 0)$.
- Their difference $\tilde{y} - \tilde{x}_0 = (-1, 1, 0)$ has entries summing to $-1 + 1 + 0 = 0$, so it is in $\mathcal{H}_0$, and $\tilde{y} = \tilde{x}_0 + (-1, 1, 0)$, as part 2 says.

•

$$\begin{aligned}\tilde{w} \cdot (-1, 1, 0) &= -1 + 1 + 0 \\ &= 0,\end{aligned}$$

as part 3 says.

- The point built in the proof is

$$\begin{aligned}\tilde{x}^* &= -\frac{1}{3} (1, 1, 1) \\ &= \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),\end{aligned}$$

$$\text{and } \frac{1}{3} + \frac{1}{3} + \frac{1}{3} - 1 = 0.$$

Theorem 0.21 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$). *Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane (Definition 19 in Matrices).*

1. $\mathcal{H}$ contains the origin $\tilde{0}$ exactly when $b = 0$.

2. If $p = 2$, then $\mathcal{H}$ is a line: $\mathcal{H} = \{\tilde{x}_0 + t\tilde{v} : t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^2$ and some nonzero $\tilde{v} \in \mathbb{R}^2$.
3. If $p = 3$, then $\mathcal{H}$ is a plane: $\mathcal{H} = \{\tilde{x}_0 + s\tilde{v}_1 + t\tilde{v}_2 : s, t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^3$ and some linearly independent $\tilde{v}_1, \tilde{v}_2 \in \mathbb{R}^3$.

i Proof

Proof. **Part 1.** Since $\tilde{w} \cdot \tilde{0} = 0$,

$$\begin{aligned}\tilde{w} \cdot \tilde{0} + b &= 0 + b \quad (\text{the dot product with } \tilde{0} \text{ is } 0) \\ &= b, \quad (\text{Theorem ??})\end{aligned}$$

so $\tilde{0} \in \mathcal{H}$ exactly when $b = 0$.

Parts 2 and 3. By Theorem 0.20, $\mathcal{H} = \{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{H}_0\}$ for any $\tilde{x}_0 \in \mathcal{H}$, where $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ of dimension $p - 1$.

For $p = 2$, $\mathcal{H}_0$ has dimension 1, so it has a basis (Definition 0.6) with one vector $\tilde{v}$. That vector is nonzero, and $\mathcal{H}_0 = \{t\tilde{v} : t \in \mathbb{R}\}$.

For $p = 3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\tilde{v}_1$ and $\tilde{v}_2$. They are linearly independent, and $\mathcal{H}_0 = \{s\tilde{v}_1 + t\tilde{v}_2 : s, t \in \mathbb{R}\}$. $\square$

Example 0.37 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points). In $\mathbb{R}^2$, take $\tilde{w} = (1, 1)$ and $b = -1$, the line of Example 19 in Matrices. The point $\tilde{x}_0 = -\frac{b}{\tilde{w} \cdot \tilde{w}} \tilde{w}$ is on it, and the vector $\tilde{v} = (-w_2, w_1)$ is nonzero with $\tilde{w} \cdot \tilde{v} = 0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.20. In $\mathbb{R}^3$, take $\tilde{w} = (1, 1, 1)$ and $b = -1$, the plane of Example 0.36, with the independent vectors $\tilde{v}_1 = (1, -1, 0)$ and $\tilde{v}_2 = (0, 1, -1)$ in $\mathcal{H}_0$. The code evaluates $\tilde{w} \cdot \tilde{x} + b$ at the origin and at points $\tilde{x}_0 + t\tilde{v}$ and $\tilde{x}_0 + s\tilde{v}_1 + t\tilde{v}_2$.

```

level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>      origin_in_R2      origin_in_R3 origin_with_b0
#>             -1             -1             0

```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\tilde{x}_0 + t \tilde{v}$ on the line and all 25 points $\tilde{x}_0 + s \tilde{v}_1 + t \tilde{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\tilde{w} \cdot \tilde{x} + b| < 10^{-10}$.

Definition 0.14 (Half-space). Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane ([Definition 19 in Matrices](#)). The two **open half-spaces** of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b < 0\}. \quad (1)$$

The **closed half-spaces** also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$. Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\tilde{w} \cdot \tilde{x} + b$ is positive, zero or negative. The vector $\tilde{w}$ points into $\mathcal{H}^+$: the number $\tilde{w} \cdot \tilde{x} + b$ goes up when $\tilde{x}$ moves in the direction $\tilde{w}$.

Remark. Replacing $(\tilde{w}, b)$ by $(-\tilde{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\tilde{w} \cdot \tilde{x} + b$ tells you which side of the hyperplane $\tilde{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.38 (Half-spaces of a line in the plane). Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\tilde{w} = (1, 2)$ and $b = -2$ (Definition 19 in Matrices), so $\mathcal{H} = \{\tilde{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\tilde{w} \cdot \tilde{x} + b$ at three points (Definition 0.14):

- At $(0, 0)$ it equals $0 + 0 - 2 = -2 < 0$, so $(0, 0) \in \mathcal{H}^-$.
- At $(2, 0)$ it equals $2 + 0 - 2 = 0$, so $(2, 0) \in \mathcal{H}$.
- At $(2, 1)$ it equals $2 + 2 - 2 = 2 > 0$, so $(2, 1) \in \mathcal{H}^+$.

Moving from $(2, 0)$ in the direction $\tilde{w} = (1, 2)$ gives $(3, 2)$, where the value is $3 + 4 - 2 = 5 > 0$. So the direction $\tilde{w}$ leads into $\mathcal{H}^+$.

Theorem 0.22 (Differences of points in a hyperplane and in a half-space). *Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, and let $\mathcal{H}, \mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.14. For a set $S \subseteq \mathbb{R}^p$, let $D(S) \stackrel{\text{def}}{=} \{\tilde{y} - \tilde{x} : \tilde{x}, \tilde{y} \in S\}$ be the set of differences of its points.*

1. $D(\mathcal{H}) = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$, a subspace of dimension $p - 1$.
2. $D(\mathcal{H}^+) = \mathbb{R}^p$, a subspace of dimension p .

The same holds for $\mathcal{H}^-$.

Proof

Proof. Part 1. Let $\tilde{x}, \tilde{y} \in \mathcal{H}$. By Theorem 0.20 part 3, $\tilde{w} \cdot (\tilde{y} - \tilde{x}) = 0$, so $\tilde{y} - \tilde{x} \in \mathcal{H}_0 = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$. Conversely, fix any $\tilde{x}_0 \in \mathcal{H}$ (it exists by Theorem 0.20 part 2). For any $\tilde{s} \in \mathcal{H}_0$, the point $\tilde{x}_0 + \tilde{s}$ is in $\mathcal{H}$ by the same part, and $(\tilde{x}_0 + \tilde{s}) - \tilde{x}_0 = \tilde{s}$. So $D(\mathcal{H}) = \mathcal{H}_0$, which has dimension $p - 1$ by Theorem 0.20 part 1.

Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+) \subseteq \mathbb{R}^p$. For the other direction, fix any $\tilde{x}_0 \in \mathcal{H}$ and any $\tilde{v} \in \mathbb{R}^p$. For a number $t > 0$ to be chosen, let $\tilde{x}_t \stackrel{\text{def}}{=} \tilde{x}_0 + t\tilde{w}$. Then

$$\begin{aligned} \tilde{w} \cdot \tilde{x}_t + b &= (\tilde{w} \cdot \tilde{x}_0 + b) + t\tilde{w} \cdot \tilde{w} \quad (\text{linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\tilde{w} \cdot \tilde{w}, \quad (\tilde{x}_0 \in \mathcal{H}) \end{aligned}$$

which is positive, so $\tilde{x}_t \in \mathcal{H}^+$. Next,

$$\begin{aligned} \tilde{w} \cdot (\tilde{x}_t + \tilde{v}) + b &= (\tilde{w} \cdot \tilde{x}_t + b) + \tilde{w} \cdot \tilde{v} \quad (\text{linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\tilde{w} \cdot \tilde{w} + \tilde{w} \cdot \tilde{v}. \quad (\text{the display above}) \end{aligned}$$

Because $\tilde{w} \cdot \tilde{w} > 0$, this is positive once $t > -\tilde{w} \cdot \tilde{v} / \tilde{w} \cdot \tilde{w}$. Choose such a $t > 0$. Then $\tilde{x}_t$ and $\tilde{x}_t + \tilde{v}$ are both in $\mathcal{H}^+$, and their difference is $\tilde{v}$. So $\tilde{v} \in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 0.8).

The proof for $\mathcal{H}^-$ is the same with $-\tilde{w}$ and $-b$ in place of $\tilde{w}$ and b . $\square$

Example 0.39 (Two iris species on either side of a line). The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species `versicolor` and `virginica`, and let $\tilde{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\tilde{x}$ gives a coefficient vector $\tilde{w}$ and an intercept b . The line $\tilde{w} \cdot \tilde{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 19 in Matrices), and the two species mostly fall in its two open half-spaces (Definition 0.14). Figure 1 shows the line and the flowers.

```
two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")
```

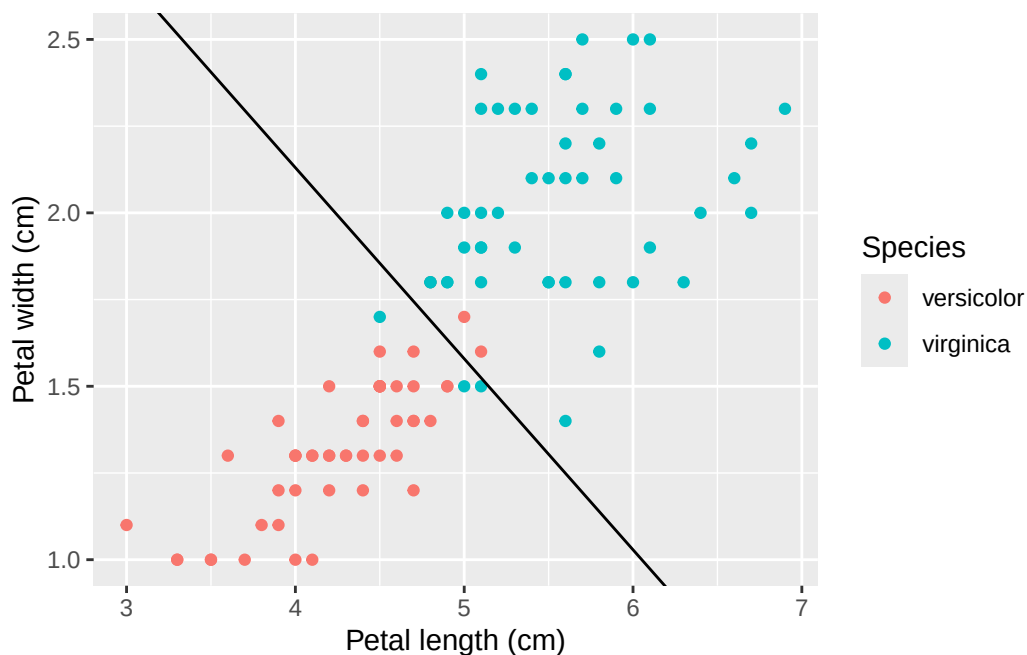


Figure 1: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>           negative positive
#> versicolor      47       3
#> virginica       3      47
```

The vector $\tilde{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p - 1 = 1$, and each side is a region of the plane of dimension $p = 2$ (Theorem 0.22).

Exercise 0.1 (Which side of the plane?). Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\tilde{w} = (1, 1, 1)$ and $b = -1$ (Example 0.36).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.14): $(0, 0, 0)$, $(1, 0, 0)$ and $(1, 1, 1)$.
2. Write the plane and its half-spaces using $-\tilde{w}$ and $-b$. Which points from part 1 change sides?
3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.22).

Solution

Solution 0.1.

1. Compute $\tilde{w} \cdot \tilde{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
 - At $(0, 0, 0)$ it equals $0 + 0 + 0 - 1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
 - At $(1, 0, 0)$ it equals $1 + 0 + 0 - 1 = 0$, so the point is in $\mathcal{H}$.
 - At $(1, 1, 1)$ it equals $1 + 1 + 1 - 1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\tilde{w} = (-1, -1, -1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0, 0, 0)$ is now in the positive half-space, and $(1, 1, 1)$ is now in the negative half-space. The point $(1, 0, 0)$ stays on the plane. So the origin and $(1, 1, 1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\tilde{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.15 (Rank factorization). Let $\mathbf{A}$ be an $m \times n$ matrix with $\text{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** of $\mathbf{A}$ is a product

$$\underbrace{\mathbf{A}}_{m \times n} = \underbrace{\mathbf{C}}_{m \times r} \underbrace{\mathbf{R}}_{r \times n}.$$

Example 0.40 (A rank factorization of a rank-one matrix). The matrix $\mathbf{A}$ in Example 0.22 has rank 1 (Example 0.29), and

$$\begin{aligned} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \end{bmatrix} &= \begin{bmatrix} 1 \cdot 1 & 1 \cdot (-2) & 1 \cdot (-2) \\ 3 \cdot 1 & 3 \cdot (-2) & 3 \cdot (-2) \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}, \end{aligned}$$

so $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} \frac{1}{2} & -1 & -1 \end{bmatrix}$ give the same product. A product $\mathbf{C}\mathbf{R}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\text{rank}(\mathbf{A}) = 1$.

Theorem 0.23 (Every nonzero matrix has a rank factorization). *Every $m \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.15). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

i Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.16), so it has a basis $\tilde{c}_1, \dots, \tilde{c}_r$ (Theorem 0.9); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\tilde{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\tilde{a}_j = r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r$ for some numbers r_{ij} , because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij} , whose column j is $\tilde{r}_j = (r_{1j}, \dots, r_{rj})$. Then column j of $\mathbf{CR}$ is

$$\begin{aligned} \mathbf{C}\tilde{r}_j &= r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r \quad (\text{Theorem 0.3}) \\ &= \tilde{a}_j, \quad (\text{choice of the } r_{ij}) \end{aligned}$$

so $\mathbf{CR} = \mathbf{A}$. □

Example 0.41 (Building a rank factorization from a basis). For $\mathbf{A}$ in Example 0.22, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (1), and it is linearly independent because it is nonzero ($c(1, 3) = \tilde{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$, $(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.23 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.40.

Sums and direct sums

i Source

This section, Section and Section are adapted from Zhou (2024a), used under the MIT License (see the license text in Section). The source proves that a subspace and its orthogonal complement together make up $\mathbb{R}^p$ by extending an orthonormal basis; the proof here uses the rank-nullity theorem (Theorem 0.17) instead.

Definition 0.16 (Sum of subspaces). The **sum** of two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$ of $\mathbb{R}^p$ (Definition 0.4) is the set of all sums of a vector from $\mathcal{S}_1$ and a vector from $\mathcal{S}_2$:

$$\mathcal{S}_1 + \mathcal{S}_2 \stackrel{\text{def}}{=} \{\tilde{x}_1 + \tilde{x}_2 : \tilde{x}_1 \in \mathcal{S}_1, \tilde{x}_2 \in \mathcal{S}_2\}.$$

Example 0.42 (Two lines sum to a plane). Let $\mathcal{S}_1 = \text{span}\{(1, 0, 0)\}$ and $\mathcal{S}_2 = \text{span}\{(0, 1, 0)\}$, two lines in $\mathbb{R}^3$. A vector of $\mathcal{S}_1 + \mathcal{S}_2$ has the form $a(1, 0, 0) + b(0, 1, 0) = (a, b, 0)$, and a and b can be any numbers, so $\mathcal{S}_1 + \mathcal{S}_2 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$.

Example 0.43 (The sum is not the union). For the two lines in Example 0.42, the union $\mathcal{S}_1 \cup \mathcal{S}_2$ contains $(1, 0, 0)$ and $(0, 1, 0)$ but not their sum $(1, 1, 0)$: every multiple of $(1, 0, 0)$ has second entry 0, and every multiple of $(0, 1, 0)$ has first entry 0. So the union is not closed under addition and is not a subspace, while the sum, the whole plane $z = 0$, is.

Theorem 0.24 (Sums and intersections of subspaces are subspaces). *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$ (Definition 0.4), then $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 0.16) and $\mathcal{S}_1 \cap \mathcal{S}_2$ are subspaces of $\mathbb{R}^p$.*

Proof

Proof. Both $\mathcal{S}_1$ and $\mathcal{S}_2$ contain $\tilde{0}$ (Theorem 0.1).

The sum. It is not empty: it contains $\tilde{0} + \tilde{0} = \tilde{0}$. Take two of its vectors, $\tilde{u} = \tilde{u}_1 + \tilde{u}_2$ and $\tilde{w} = \tilde{w}_1 + \tilde{w}_2$, with $\tilde{u}_1, \tilde{w}_1 \in \mathcal{S}_1$ and $\tilde{u}_2, \tilde{w}_2 \in \mathcal{S}_2$, and a number c . Then

$$\begin{aligned}\tilde{u} + \tilde{w} &= (\tilde{u}_1 + \tilde{u}_2) + (\tilde{w}_1 + \tilde{w}_2) \quad (\text{substitute}) \\ &= (\tilde{u}_1 + \tilde{w}_1) + (\tilde{u}_2 + \tilde{w}_2), \quad (\text{regroup; vector addition is entrywise})\end{aligned}$$

where $\tilde{u}_1 + \tilde{w}_1 \in \mathcal{S}_1$ and $\tilde{u}_2 + \tilde{w}_2 \in \mathcal{S}_2$, because each subspace is closed under addition. Likewise

$$\begin{aligned}c\tilde{u} &= c(\tilde{u}_1 + \tilde{u}_2) \quad (\text{substitute}) \\ &= c\tilde{u}_1 + c\tilde{u}_2, \quad (\text{distribute } c \text{ entrywise})\end{aligned}$$

where $c\tilde{u}_1 \in \mathcal{S}_1$ and $c\tilde{u}_2 \in \mathcal{S}_2$, because each subspace is closed under scalar multiplication. So $\tilde{u} + \tilde{w}$ and $c\tilde{u}$ are in $\mathcal{S}_1 + \mathcal{S}_2$.

The intersection. It is not empty: it contains $\tilde{0}$. If $\tilde{u}$ and $\tilde{w}$ are in both $\mathcal{S}_1$ and $\mathcal{S}_2$, then $\tilde{u} + \tilde{w}$ and $c\tilde{u}$ are in $\mathcal{S}_1$, because $\mathcal{S}_1$ is a subspace, and in $\mathcal{S}_2$ for the same reason, so they are in $\mathcal{S}_1 \cap \mathcal{S}_2$. $\square$

Example 0.44 (Two planes in $\mathbb{R}^3$). Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0, b, c) : b, c \in \mathbb{R}\}$, the plane $x = 0$.

- **Intersection:** a vector in both has third entry 0 and first entry 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{(0, b, 0) : b \in \mathbb{R}\}$, the y -axis, which is a line through the origin.
- **Sum:** $(a, b, 0) + (0, b', c) = (a, b + b', c)$, and any (x, y, z) arises this way, with $a = x$,

$b = y$, $b' = 0$ and $c = z$. So $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.

Theorem 0.25 (The dimension of a sum of subspaces). *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2) - \dim(\mathcal{S}_1 \cap \mathcal{S}_2).$$

i Proof

Proof. All three sets in the formula are subspaces (Theorem 0.24), so each has a dimension (Definition 0.8). Let $k = \dim(\mathcal{S}_1 \cap \mathcal{S}_2)$, $d_1 = \dim(\mathcal{S}_1)$ and $d_2 = \dim(\mathcal{S}_2)$.

Build a list. Choose a basis $\tilde{z}_1, \dots, \tilde{z}_k$ of $\mathcal{S}_1 \cap \mathcal{S}_2$ (Theorem 0.9). These vectors are linearly independent and lie in $\mathcal{S}_1$, so Theorem 0.9 extends them to a basis $\tilde{z}_1, \dots, \tilde{z}_k, \tilde{x}_1, \dots, \tilde{x}_{d_1-k}$ of $\mathcal{S}_1$, which has d_1 vectors (Theorem 0.7). In the same way, extend them to a basis $\tilde{z}_1, \dots, \tilde{z}_k, \tilde{y}_1, \dots, \tilde{y}_{d_2-k}$ of $\mathcal{S}_2$. We show that the $\tilde{z}$'s, $\tilde{x}$'s and $\tilde{y}$'s together, $k + (d_1 - k) + (d_2 - k) = d_1 + d_2 - k$ vectors, are a basis of $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 0.6).

They span $\mathcal{S}_1 + \mathcal{S}_2$. Any $\tilde{v} \in \mathcal{S}_1 + \mathcal{S}_2$ is $\tilde{v}_1 + \tilde{v}_2$ with $\tilde{v}_1 \in \mathcal{S}_1$ and $\tilde{v}_2 \in \mathcal{S}_2$. $\tilde{v}_1$ is a linear combination of the $\tilde{z}$'s and $\tilde{x}$'s, and $\tilde{v}_2$ is a linear combination of the $\tilde{z}$'s and $\tilde{y}$'s, so their sum is a linear combination of all three groups. Conversely, each vector in the list is in $\mathcal{S}_1 + \mathcal{S}_2$ (for example, $\tilde{x}_1 = \tilde{x}_1 + \tilde{0}$), and $\mathcal{S}_1 + \mathcal{S}_2$ is a subspace, so it contains every linear combination of them.

They are linearly independent. Suppose

$$\sum_{i=1}^k a_i \tilde{z}_i + \sum_{j=1}^{d_1-k} b_j \tilde{x}_j + \sum_{l=1}^{d_2-k} c_l \tilde{y}_l = \tilde{0}.$$

Let $\tilde{w} \stackrel{\text{def}}{=} \sum_l c_l \tilde{y}_l$. Then $\tilde{w} \in \mathcal{S}_2$, because it is a linear combination of vectors of $\mathcal{S}_2$. Subtracting $\sum_i a_i \tilde{z}_i + \sum_j b_j \tilde{x}_j$ from both sides of the supposed equation gives

$$\tilde{w} = \sum_{i=1}^k (-a_i) \tilde{z}_i + \sum_{j=1}^{d_1-k} (-b_j) \tilde{x}_j,$$

a linear combination of vectors of $\mathcal{S}_1$, so $\tilde{w} \in \mathcal{S}_1$ too. So $\tilde{w} \in \mathcal{S}_1 \cap \mathcal{S}_2$, and $\tilde{w} = \sum_i e_i \tilde{z}_i$ for some numbers e_i , since the $\tilde{z}$'s span the intersection. Then

$$\begin{aligned} \tilde{0} &= \tilde{w} - \tilde{w} && \text{(a vector minus itself)} \\ &= \sum_{i=1}^k e_i \tilde{z}_i - \sum_{l=1}^{d_2-k} c_l \tilde{y}_l && \text{(substitute the two expressions for } \tilde{w}) \\ &= \sum_{i=1}^k e_i \tilde{z}_i + \sum_{l=1}^{d_2-k} (-c_l) \tilde{y}_l, && \text{(write each subtraction as adding } -1 \text{ times)} \end{aligned}$$

and because the $\tilde{z}$'s and $\tilde{y}$'s are a basis of $\mathcal{S}_2$, every $c_l = 0$. The supposed equation then reads $\sum_i a_i \tilde{z}_i + \sum_j b_j \tilde{x}_j = \tilde{0}$, and because the $\tilde{z}$'s and $\tilde{x}$'s are a basis of $\mathcal{S}_1$, every a_i and b_j is 0.

So $\dim(\mathcal{S}_1 + \mathcal{S}_2) = d_1 + d_2 - k$. When some of the groups are empty (for example $k = 0$, or $\mathcal{S}_1 = \mathcal{S}_1 \cap \mathcal{S}_2$), the sums over them are $\tilde{0}$ and the argument goes through unchanged. $\square$

Example 0.45 (Two planes in $\mathbb{R}^3$, by dimension). For the planes $z = 0$ and $x = 0$ in Example 0.44, each has dimension 2: the plane $z = 0$ has basis $(1, 0, 0), (0, 1, 0)$ (Example 0.16), and the plane $x = 0$ has basis $(0, 1, 0), (0, 0, 1)$ by the same argument. Their intersection, the y -axis $\text{span}\{(0, 1, 0)\}$, has dimension 1, because the single nonzero vector $(0, 1, 0)$ is linearly independent ($c(0, 1, 0) = (0, c, 0)$ is $\tilde{0}$ only if $c = 0$) and so is a basis of it. Theorem 0.25 gives

$$\begin{aligned}\dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 + 2 - 1 \\ &= 3,\end{aligned}$$

which agrees with $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$ (Example 0.44) and $\dim(\mathbb{R}^3) = 3$ (Example 0.16).

Corollary 0.2 (The dimension of a sum is at most the sum of the dimensions). *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) \leq \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2),$$

with equality exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$.

i Proof

Proof. By Theorem 0.25, the two sides differ by $\dim(\mathcal{S}_1 \cap \mathcal{S}_2) \geq 0$. That dimension is 0 exactly when a basis of the intersection is the empty list, that is, when the intersection is the span of the empty list, $\{\tilde{0}\}$ (Definition 0.5, Example 0.16). $\square$

Example 0.46 (Equality and strict inequality).

- For the lines $\mathcal{S}_1 = \text{span}\{(1, 0, 0)\}$ and $\mathcal{S}_2 = \text{span}\{(0, 1, 0)\}$ in Example 0.42, a common vector satisfies $a(1, 0, 0) = b(0, 1, 0)$, so $(a, -b, 0) = \tilde{0}$ and

$$\begin{aligned}a &= b \\ &= 0.\end{aligned}$$

The intersection is $\{\tilde{0}\}$, and indeed

$$\begin{aligned}\dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 \\ &= 1 + 1 : \end{aligned}$$

the sum is the plane $z = 0$ (Example 0.42), which has dimension 2 (Example 0.16), and each line has dimension 1, with its spanning vector as a basis.

- For

$$\begin{aligned}\mathcal{S}_1 &= \mathcal{S}_2 \\ &= \text{span}\{(1, 0, 0)\},\end{aligned}$$

the sum is the same line, because $a(1, 0, 0) + b(1, 0, 0) = (a+b)(1, 0, 0)$, so $\dim(\mathcal{S}_1 + \mathcal{S}_2) = 1 < 1 + 1$, a [strict inequality](#).

Theorem 0.26 (Rank is subadditive). *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$,*

$$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

i Proof

Proof. $\mathcal{C}(\mathbf{A} + \mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$. Any vector of $\mathcal{C}(\mathbf{A} + \mathbf{B})$ is $(\mathbf{A} + \mathbf{B})\tilde{x}$ for some $\tilde{x} \in \mathbb{R}^n$ (Definition 0.9), and $(\mathbf{A} + \mathbf{B})\tilde{x} = \mathbf{A}\tilde{x} + \mathbf{B}\tilde{x}$ (Theorem 6 in Matrices, with $\tilde{x}$ as an $n \times 1$ matrix), a vector of $\mathcal{C}(\mathbf{A})$ plus a vector of $\mathcal{C}(\mathbf{B})$.

Compare dimensions. All the sets involved are subspaces of $\mathbb{R}^m$ (Theorem 0.12, Theorem 0.24), so

$$\begin{aligned}\text{rank}(\mathbf{A} + \mathbf{B}) &= \dim(\mathcal{C}(\mathbf{A} + \mathbf{B})) && \text{(Theorem 0.16)} \\ &\leq \dim(\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})) && \text{(the containment, and part 2 of Theorem 0.10)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) + \dim(\mathcal{C}(\mathbf{B})) && \text{(Corollary 0.2)} \\ &= \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}). && \text{(Theorem 0.16)}\end{aligned}$$

□

Example 0.47 (Subadditivity with equality and without). Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Each has one nonzero column and one zero column. The nonzero column on its own is linearly independent ($c\tilde{v} = \tilde{0}_2$ with $\tilde{v} \neq \tilde{0}_2$ forces $c = 0$), and the two columns together are not, since 1 times the zero column is $\tilde{0}_2$; so each matrix has rank 1

(Definition 0.2). By the same argument, $-\mathbf{A}$ has rank 1.

- $\mathbf{A} + \mathbf{B}$ is the 2×2 identity matrix, which has rank 2 (Example 0.31), so the bound $2 \leq 1 + 1$ holds with equality.
- $\mathbf{A} + (-\mathbf{A})$ is the 2×2 zero matrix, which has rank 0: any nonempty list of its columns contains $\tilde{0}_2$, and $1 \cdot \tilde{0}_2 = \tilde{0}_2$. So the bound $0 \leq \text{rank}(\mathbf{A}) + \text{rank}(-\mathbf{A}) = 1 + 1$ is strict.

Definition 0.17 (Direct sum). Let $\mathcal{S}_1$, $\mathcal{S}_2$ and $\mathcal{V}$ be subspaces of $\mathbb{R}^p$. $\mathcal{S}_1$ and $\mathcal{S}_2$ are **complementary** in $\mathcal{V}$ if

$$\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2 \quad \text{and} \quad \mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$$

(Definition 0.16). Then $\mathcal{V}$ is the **direct sum** of $\mathcal{S}_1$ and $\mathcal{S}_2$, written $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.48 ($\mathbb{R}^3$ is a plane plus a line). Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0, 0, c) : c \in \mathbb{R}\}$, the z -axis.

- **Sum:** any $(x, y, z) \in \mathbb{R}^3$ is $(x, y, 0) + (0, 0, z)$, so $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.
- **Intersection:** a vector in both has third entry 0 and first and second entries 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$.

So $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.49 (Two planes that sum to $\mathbb{R}^3$ but not directly). The planes $z = 0$ and $x = 0$ sum to $\mathbb{R}^3$ (Example 0.44), but they share the y -axis, so they are not complementary, and $\mathbb{R}^3$ is not their direct sum. The shared line shows up as more than one way to split a vector:

$$\begin{aligned} (0, 1, 0) &= (0, 1, 0) + \tilde{0} \\ &= \tilde{0} + (0, 1, 0), \end{aligned}$$

with the first term in the plane $z = 0$ and the second in the plane $x = 0$ both times.

Theorem 0.27 (Three ways to recognize a direct sum). *Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be subspaces of $\mathbb{R}^p$, and let $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. The following statements are equivalent:*

1. $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$ (Definition 0.17).
2. $\dim(\mathcal{V}) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$.
3. Every $\tilde{x} \in \mathcal{V}$ can be written as $\tilde{x} = \tilde{x}_1 + \tilde{x}_2$ with $\tilde{x}_1 \in \mathcal{S}_1$ and $\tilde{x}_2 \in \mathcal{S}_2$ in only one way.

i Proof

Proof. Since $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$ is given, statement 1 says exactly that $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$.

1 and 2 are equivalent. By Corollary 0.2, $\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$ exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$.

1 implies 3. Every $\tilde{x} \in \mathcal{V}$ has at least one such expression, because $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. Suppose

$$\begin{aligned}\tilde{x} &= \tilde{u}_1 + \tilde{u}_2 \\ &= \tilde{v}_1 + \tilde{v}_2,\end{aligned}$$

with $\tilde{u}_1, \tilde{v}_1 \in \mathcal{S}_1$ and $\tilde{u}_2, \tilde{v}_2 \in \mathcal{S}_2$. Then

$$\begin{aligned}\tilde{u}_1 - \tilde{v}_1 &= (\tilde{u}_1 + \tilde{u}_2) - \tilde{u}_2 - \tilde{v}_1 && \text{(add and subtract } \tilde{u}_2) \\ &= (\tilde{v}_1 + \tilde{v}_2) - \tilde{u}_2 - \tilde{v}_1 && \text{(substitute } \tilde{u}_1 + \tilde{u}_2 = \tilde{v}_1 + \tilde{v}_2) \\ &= \tilde{v}_2 - \tilde{u}_2. && \text{(cancel } \tilde{v}_1)\end{aligned}$$

The left side is in $\mathcal{S}_1$ and the right side is in $\mathcal{S}_2$, because each subspace is closed under addition and scalar multiplication (the difference is $\tilde{u}_1 + (-1)\tilde{v}_1$). So both sides are in $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$, which gives $\tilde{u}_1 = \tilde{v}_1$ and $\tilde{u}_2 = \tilde{v}_2$.

3 implies 1. Take any $\tilde{x} \in \mathcal{S}_1 \cap \mathcal{S}_2$. Then $\tilde{x} = \tilde{x} + \tilde{0}$, with $\tilde{x} \in \mathcal{S}_1$ and $\tilde{0} \in \mathcal{S}_2$, and $\tilde{x} = \tilde{0} + \tilde{x}$, with $\tilde{0} \in \mathcal{S}_1$ and $\tilde{x} \in \mathcal{S}_2$ (Theorem 0.1). By statement 3 these two expressions are the same, so $\tilde{x} = \tilde{0}$, and $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$. $\square$

Example 0.50 (Checking the three statements). For the plane $z = 0$ and the z -axis in Example 0.48:

- the dimensions add up:

$$\begin{aligned}2 + 1 &= 3 \\ &= \dim(\mathbb{R}^3),\end{aligned}$$

since the plane has dimension 2 (Example 0.16), the z -axis has the single nonzero vector $(0, 0, 1)$ as a basis, and $\dim(\mathbb{R}^3) = 3$ (Example 0.16);

- the split is unique: if

$$\begin{aligned}(x, y, z) &= (a, b, 0) + (0, 0, c) \\ &= (a, b, c),\end{aligned}$$

then $a = x$, $b = y$ and $c = z$.

For the planes $z = 0$ and $x = 0$ in Example 0.49, both statements fail:

$$\begin{aligned}
2 + 2 &= 4 \\
&\neq 3 \\
&= \dim(\mathcal{S}_1 + \mathcal{S}_2)
\end{aligned}$$

(Example 0.45), and $(0, 1, 0)$ splits in two ways.

Definition 0.18 (Projection onto a subspace along a complement). Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathbb{R}^p$ with $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{T}$ (Definition 0.17), and let $\tilde{y} \in \mathbb{R}^p$. By Theorem 0.27, $\tilde{y} = \tilde{u} + \tilde{v}$ for exactly one $\tilde{u} \in \mathcal{S}$ and $\tilde{v} \in \mathcal{T}$. The vector $\tilde{u}$ is the **projection** of $\tilde{y}$ onto $\mathcal{S}$ along $\mathcal{T}$ (Banerjee and Roy 2014, Definition 6.3, p. 163).

Example 0.51 (The projection depends on the complement). Let $\mathcal{S} = \text{span}\{(1, 0)\}$, the horizontal axis in $\mathbb{R}^2$, and let $\tilde{y} = (3, 2)$.

- **Along $\mathcal{T}_1 = \text{span}\{(1, -1)\}$:** $(3, 2) = (5, 0) + (-2, 2)$, with $(5, 0) \in \mathcal{S}$ and $(-2, 2) = -2 \cdot (1, -1) \in \mathcal{T}_1$, so the projection of $(3, 2)$ onto $\mathcal{S}$ along $\mathcal{T}_1$ is $(5, 0)$.
- **Along $\mathcal{T}_2 = \text{span}\{(0, 1)\}$:** $(3, 2) = (3, 0) + (0, 2)$, with $(3, 0) \in \mathcal{S}$ and $(0, 2) \in \mathcal{T}_2$, so the projection of $(3, 2)$ onto $\mathcal{S}$ along $\mathcal{T}_2$ is $(3, 0)$.

Each of $\mathcal{T}_1$ and $\mathcal{T}_2$ is a line that meets $\mathcal{S}$ only at $\tilde{0}$, so

$$\begin{aligned}
\mathbb{R}^2 &= \mathcal{S} \oplus \mathcal{T}_1 \\
&= \mathcal{S} \oplus \mathcal{T}_2
\end{aligned}$$

(Definition 0.17, Theorem 0.27). The same vector has a different projection onto the same subspace when the complement changes.

Orthogonal complements

Theorem 0.28 (The dot product is linear in each slot). *For vectors $\tilde{x}, \tilde{u}, \tilde{w} \in \mathbb{R}^p$ and numbers a, b ,*

$$\tilde{x} \cdot (a \tilde{u} + b \tilde{w}) = a (\tilde{x} \cdot \tilde{u}) + b (\tilde{x} \cdot \tilde{w})$$

and

$$(a \tilde{u} + b \tilde{w}) \cdot \tilde{x} = a (\tilde{u} \cdot \tilde{x}) + b (\tilde{w} \cdot \tilde{x}).$$

i Proof

Proof. **Second slot.**

$$\begin{aligned}
 \tilde{x} \cdot (a \tilde{u} + b \tilde{w}) &= \sum_{i=1}^p x_i (a \tilde{u}_i + b \tilde{w}_i) && \text{(Definition 7 in Vectors)} \\
 &= \sum_{i=1}^p x_i (a u_i + b w_i) && \text{(Definition 6 in Matrices, Definition 6 in Vectors)} \\
 &= \sum_{i=1}^p (x_i a u_i + x_i b w_i) && \text{(distribute each } x_i) \\
 &= \sum_{i=1}^p (a x_i u_i + b x_i w_i) && \text{(commute the factors in each product)} \\
 &= \sum_{i=1}^p a x_i u_i + \sum_{i=1}^p b x_i w_i && \text{(split the finite sum)} \\
 &= a \sum_{i=1}^p x_i u_i + b \sum_{i=1}^p x_i w_i && \text{(factor } a \text{ and } b \text{ out of the sums)} \\
 &= a (\tilde{x} \cdot \tilde{u}) + b (\tilde{x} \cdot \tilde{w}). && \text{(Definition 7 in Vectors)}
 \end{aligned}$$

First slot.

$$\begin{aligned}
 (a \tilde{u} + b \tilde{w}) \cdot \tilde{x} &= \tilde{x} \cdot (a \tilde{u} + b \tilde{w}) && \text{(Theorem 1 in Vectors)} \\
 &= a (\tilde{x} \cdot \tilde{u}) + b (\tilde{x} \cdot \tilde{w}) && \text{(second slot)} \\
 &= a (\tilde{u} \cdot \tilde{x}) + b (\tilde{w} \cdot \tilde{x}). && \text{(Theorem 1 in Vectors, twice)}
 \end{aligned}$$

Two special cases are used often. With

$$\begin{aligned}
 a &= b \\
 &= 1,
 \end{aligned}$$

and $1 \tilde{u} = \tilde{u}$, the theorem gives $\tilde{x} \cdot (\tilde{u} + \tilde{w}) = \tilde{x} \cdot \tilde{u} + \tilde{x} \cdot \tilde{w}$. With $\tilde{w} = \tilde{u}$ and $b = 0$, and $0 \tilde{u} = \tilde{0}$, it gives $\tilde{x} \cdot (a \tilde{u}) = a (\tilde{x} \cdot \tilde{u})$, and likewise in the first slot. $\square$

Example 0.52 (Splitting a dot product). Let $\tilde{x} = (1, 2)$, $\tilde{u} = (3, 0)$, $\tilde{w} = (0, 1)$, $a = 2$ and $b = -1$. Directly, $a \tilde{u} + b \tilde{w} = (6, -1)$ and

$$\begin{aligned}
 \tilde{x} \cdot (6, -1) &= 6 - 2 \\
 &= 4.
 \end{aligned}$$

By Theorem 0.28,

$$\begin{aligned} a(\tilde{x} \cdot \tilde{u}) + b(\tilde{x} \cdot \tilde{w}) &= 2 \cdot 3 + (-1) \cdot 2 \\ &= 4, \end{aligned}$$

the same number.

Definition 0.19 (Orthogonal complement). The **orthogonal complement** of a set $\mathcal{X}$ of vectors in $\mathbb{R}^p$ is the set of vectors orthogonal ([Definition 13 in Vectors](#)) to every vector of $\mathcal{X}$:

$$\mathcal{X}^\perp \stackrel{\text{def}}{=} \{\tilde{u} \in \mathbb{R}^p : \tilde{x} \cdot \tilde{u} = 0 \text{ for all } \tilde{x} \in \mathcal{X}\}.$$

$\mathcal{X}^\perp$ is read “ $\mathcal{X}$ perp”. $\mathcal{X}$ need not be a subspace.

Example 0.53 (Orthogonal complements in $\mathbb{R}^3$).

- **A plane.** Let $\mathcal{X} = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$. A vector $\tilde{u}$ is in $\mathcal{X}^\perp$ when

$$\begin{aligned} (a, b, 0) \cdot \tilde{u} &= au_1 + bu_2 \\ &= 0 \end{aligned}$$

for all a and b . Taking $(a, b) = (1, 0)$ forces $u_1 = 0$, and taking $(a, b) = (0, 1)$ forces $u_2 = 0$; conversely, if

$$\begin{aligned} u_1 &= u_2 \\ &= 0, \end{aligned}$$

then $au_1 + bu_2 = 0$ for all a and b . So $\mathcal{X}^\perp = \{(0, 0, c) : c \in \mathbb{R}\}$, the z -axis.

- **Two vectors.** The two-element set $\{(1, 0, 0), (0, 1, 0)\}$ is not a subspace, and its orthogonal complement is the z -axis too: the two conditions $u_1 = 0$ and $u_2 = 0$ are the same as for the plane.
- **The extremes.** $\{\tilde{0}\}^\perp = \mathbb{R}^3$, because $\tilde{0} \cdot \tilde{u} = 0$ for every $\tilde{u}$. $(\mathbb{R}^3)^\perp = \{\tilde{0}\}$, because a $\tilde{u}$ in it is orthogonal to itself, and

$$\begin{aligned} \tilde{u} \cdot \tilde{u} &= u_1^2 + u_2^2 + u_3^2 \\ &= 0 \end{aligned}$$

forces $\tilde{u} = \tilde{0}$.

Theorem 0.29 (An orthogonal complement is a subspace). *For any set $\mathcal{X}$ of vectors in $\mathbb{R}^p$, $\mathcal{X}^\perp$ (Definition 0.19) is a subspace of $\mathbb{R}^p$ (Definition 0.4).*

i Proof

Proof. $\mathcal{X}^\perp$ is not empty:

$$\begin{aligned}\tilde{x} \cdot \tilde{0} &= \sum_i x_i \cdot 0 \\ &= 0\end{aligned}$$

for every $\tilde{x}$, so $\tilde{0} \in \mathcal{X}^\perp$. Let $\tilde{u}, \tilde{w} \in \mathcal{X}^\perp$, let $c \in \mathbb{R}$, and let $\tilde{x} \in \mathcal{X}$.

Addition.

$$\begin{aligned}\tilde{x} \cdot (\tilde{u} + \tilde{w}) &= \tilde{x} \cdot \tilde{u} + \tilde{x} \cdot \tilde{w} \quad (\text{Theorem 0.28, special case } a \text{ and } b \text{ both equal to } 1) \\ &= 0 + 0 \quad (\tilde{u}, \tilde{w} \in \mathcal{X}^\perp) \\ &= 0. \quad (\text{arithmetic})\end{aligned}$$

Scalar multiplication.

$$\begin{aligned}\tilde{x} \cdot (c \tilde{u}) &= c (\tilde{x} \cdot \tilde{u}) \quad (\text{Theorem 0.28, special case } b = 0) \\ &= c \cdot 0 \quad (\tilde{u} \in \mathcal{X}^\perp) \\ &= 0. \quad (\text{arithmetic})\end{aligned}$$

These two equations hold for every $\tilde{x} \in \mathcal{X}$, so $\tilde{u} + \tilde{w}$ and $c \tilde{u}$ are in $\mathcal{X}^\perp$. □

Example 0.54 (The complement of a single vector is a line). The one-element set $\mathcal{X} = \{(1, 2)\}$ in $\mathbb{R}^2$ is not a subspace: $2(1, 2) = (2, 4)$ is not in it. Its orthogonal complement is

$$\begin{aligned}\mathcal{X}^\perp &= \{\tilde{u} \in \mathbb{R}^2 : u_1 + 2u_2 = 0\} \\ &= \{c(-2, 1) : c \in \mathbb{R}\},\end{aligned}$$

since $u_1 + 2u_2 = 0$ means

$$\begin{aligned}\tilde{u} &= (-2u_2, u_2) \\ &= u_2(-2, 1).\end{aligned}$$

That set is $\text{span}\{(-2, 1)\}$, a subspace by Theorem 0.2, as Theorem 0.29 says it must be. It contains $(-2, 1)$, the vector [Remark 7 in Vectors](#) found perpendicular to $(1, 2)$.

Theorem 0.30 (The orthogonal complement of a column space is a null space). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top).$$

i Proof

Proof. Let $\tilde{a}_1, \dots, \tilde{a}_n$ be the columns of $\mathbf{A}$, so $\tilde{a}_j = (a_{1j}, \dots, a_{mj})$. For $\tilde{x} \in \mathbb{R}^m$, entry j of $\mathbf{A}^\top \tilde{x}$ is

$$\begin{aligned} (\mathbf{A}^\top \tilde{x})_j &= \sum_{i=1}^m (\mathbf{A}^\top)_{ji} x_i \quad (\text{Definition 11 in Matrices}) \\ &= \sum_{i=1}^m a_{ij} x_i \quad (\text{Definition 3 in Matrices}) \\ &= \tilde{a}_j \cdot \tilde{x}. \quad (\text{Definition 7 in Vectors}) \end{aligned}$$

So $\tilde{x} \in \mathcal{N}(\mathbf{A}^\top)$ (Definition 0.10) exactly when $\tilde{a}_j \cdot \tilde{x} = 0$ for every column $\tilde{a}_j$.
 $\mathcal{C}(\mathbf{A})^\perp \subseteq \mathcal{N}(\mathbf{A}^\top)$. Each column $\tilde{a}_j$ is in $\mathcal{C}(\mathbf{A})$ (Theorem 0.12), so a vector orthogonal to all of $\mathcal{C}(\mathbf{A})$ is orthogonal to every column.
 $\mathcal{N}(\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{A})^\perp$. Suppose $\tilde{a}_j \cdot \tilde{x} = 0$ for every j , and take any $\tilde{y} \in \mathcal{C}(\mathbf{A})$, so $\tilde{y} = \mathbf{A}\tilde{z}$ for some $\tilde{z} \in \mathbb{R}^n$ (Definition 0.9). Then

$$\begin{aligned} \tilde{y} \cdot \tilde{x} &= \sum_{i=1}^m (\mathbf{A}\tilde{z})_i x_i \quad (\text{Definition 7 in Vectors}) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} z_j \right) x_i \quad (\text{Definition 11 in Matrices}) \\ &= \sum_{i=1}^m \sum_{j=1}^n a_{ij} z_j x_i \quad (\text{distribute each } x_i \text{ over the inner sum}) \\ &= \sum_{j=1}^n \sum_{i=1}^m a_{ij} z_j x_i \quad (\text{swap the order of the finite sums}) \\ &= \sum_{j=1}^n z_j \sum_{i=1}^m a_{ij} x_i \quad (\text{commute, then factor } z_j \text{ out of the inner sum}) \\ &= \sum_{j=1}^n z_j (\tilde{a}_j \cdot \tilde{x}) \quad (\text{Definition 7 in Vectors}) \\ &= \sum_{j=1}^n z_j \cdot 0 \quad (\text{each } \tilde{a}_j \cdot \tilde{x} = 0) \\ &= 0. \quad (\text{arithmetic}) \end{aligned}$$

So $\tilde{x}$ is orthogonal to every vector of $\mathcal{C}(\mathbf{A})$. □

Example 0.55 (The complement of the column space in Example 0.22). For $\mathbf{A}$ in Example 0.22, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (1). Example 0.24 found $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$, so by Theorem 0.30 this line is $\mathcal{C}(\mathbf{A})^\perp$. Directly, for any c ,

$$\begin{aligned} (c(1, 3)) \cdot (-3, 1) &= c((1, 3) \cdot (-3, 1)) && \text{(Theorem 0.28, special case } b = 0, \text{ first slot)} \\ &= c(-3 + 3) && \text{(Definition 7 in Vectors)} \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

Theorem 0.31 (A subspace and its orthogonal complement make up the whole space). *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$. Then*

1. $\mathcal{S} \cap \mathcal{S}^\perp = \{\tilde{0}\}$;
2. $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = p$;
3. $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$ (Definition 0.17).

So every $\tilde{y} \in \mathbb{R}^p$ can be written as $\tilde{y} = \tilde{u} + \tilde{v}$ with $\tilde{u} \in \mathcal{S}$ and $\tilde{v} \in \mathcal{S}^\perp$ in only one way (Theorem 0.27).

i Proof

Proof. $\mathcal{S}^\perp$ is a subspace (Theorem 0.29), so it has a dimension (Definition 0.8).

Part 1. Both $\mathcal{S}$ and $\mathcal{S}^\perp$ contain $\tilde{0}$ (Theorem 0.1). If $\tilde{x}$ is in both, then $\tilde{x}$ is orthogonal to every vector of $\mathcal{S}$, $\tilde{x}$ itself included, so

$$\begin{aligned} \tilde{x} \cdot \tilde{x} &= x_1^2 + \cdots + x_p^2 \\ &= 0, \end{aligned}$$

which forces every $x_i = 0$.

Part 2. Let $d = \dim(\mathcal{S})$. If $d = 0$, then $\mathcal{S}$ has a basis (Theorem 0.9) with 0 vectors (Definition 0.8), the empty list, whose span is $\{\tilde{0}\}$ (Definition 0.5); so $\mathcal{S} = \{\tilde{0}\}$, $\mathcal{S}^\perp = \mathbb{R}^p$ (every $\tilde{u}$ satisfies $\tilde{0} \cdot \tilde{u} = 0$), and $0 + p = p$ (Example 0.16). If $d \geq 1$, choose a basis $\tilde{a}_1, \dots, \tilde{a}_d$ of $\mathcal{S}$ (Theorem 0.9), and let $\mathbf{A}$ be the $p \times d$ matrix with these columns. The basis spans $\mathcal{S}$, so $\mathcal{S} = \mathcal{C}(\mathbf{A})$ (Theorem 0.12), and $\mathcal{S}^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 0.30). $\mathbf{A}^\top$ is $d \times p$, so

$$\begin{aligned}
\dim(\mathcal{S}^\perp) &= \dim(\mathcal{N}(\mathbf{A}^\top)) && \text{(Theorem 0.30)} \\
&= \text{nullity}(\mathbf{A}^\top) && \text{(Definition 0.12)} \\
&= p - \text{rank}(\mathbf{A}^\top) && \text{(Theorem 0.17, for the } p \text{ columns of } \mathbf{A}^\top) \\
&= p - \text{rank}(\mathbf{A}) && \text{(Theorem 0.19)} \\
&= p - \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 0.16)} \\
&= p - d. && (\mathcal{C}(\mathbf{A}) = \mathcal{S})
\end{aligned}$$

Part 3. $\mathcal{S} + \mathcal{S}^\perp$ is a subspace of $\mathbb{R}^p$ (Theorem 0.24), and

$$\begin{aligned}
\dim(\mathcal{S} + \mathcal{S}^\perp) &= \dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(Theorem 0.25)} \\
&= p - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(part 2)} \\
&= p - 0 && \text{(part 1, and } \dim(\{\tilde{0}\}) = 0 \text{ by Example 0.16)} \\
&= p. && \text{(arithmetic)}
\end{aligned}$$

Since $\mathcal{S} + \mathcal{S}^\perp \subseteq \mathbb{R}^p$, $\mathbb{R}^p$ is a subspace (Example 0.3) and $\dim(\mathbb{R}^p) = p$ (Example 0.16), Theorem 0.11 gives $\mathcal{S} + \mathcal{S}^\perp = \mathbb{R}^p$. Together with part 1, $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$. The uniqueness of the split then follows from Theorem 0.27 (statement 1 implies statement 3). $\square$

Example 0.56 (Splitting a vector of $\mathbb{R}^3$ along a line and its complement). Let $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ in $\mathbb{R}^3$. A single nonzero vector is linearly independent ($c\tilde{v} = \tilde{0}$ with $\tilde{v} \neq \tilde{0}$ forces $c = 0$), so $(1, 1, 0)$ is a basis of $\mathcal{S}$ and $\dim(\mathcal{S}) = 1$. A vector $\tilde{u}$ is in $\mathcal{S}^\perp$ exactly when $u_1 + u_2 = 0$: that condition says $(1, 1, 0) \cdot \tilde{u} = 0$, and then

$$\begin{aligned}
c(1, 1, 0) \cdot \tilde{u} &= c(u_1 + u_2) \\
&= 0
\end{aligned}$$

for every c . So

$$\begin{aligned}
\mathcal{S}^\perp &= \{(a, -a, c) : a, c \in \mathbb{R}\} \\
&= \text{span}\{(1, -1, 0), (0, 0, 1)\}.
\end{aligned}$$

Those two vectors are linearly independent, because $a(1, -1, 0) + c(0, 0, 1) = (a, -a, c)$ is $\tilde{0}$ only if

$$\begin{aligned}
a &= c \\
&= 0,
\end{aligned}$$

so $\dim(\mathcal{S}^\perp) = 2$, and $1 + 2 = 3$, as part 2 says.

To split $\tilde{y} = (3, 1, 2)$, look for $\tilde{u} = t(1, 1, 0)$ with

$$\begin{aligned}\tilde{v} &= \tilde{y} - \tilde{u} \\ &= (3 - t, 1 - t, 2)\end{aligned}$$

in $\mathcal{S}^\perp$: that membership needs $(3 - t) + (1 - t) = 0$, so $t = 2$. Then $\tilde{u} = (2, 2, 0)$ and $\tilde{v} = (1, -1, 2)$, with $\tilde{u} + \tilde{v} = (3, 1, 2)$ and

$$\begin{aligned}(1, 1, 0) \cdot (1, -1, 2) &= 1 - 1 + 0 \\ &= 0.\end{aligned}$$

Theorem 0.32 (The orthogonal complement of the orthogonal complement). *For any subspace $\mathcal{S}$ of $\mathbb{R}^p$,*

$$(\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Proof

Proof. $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$. Take $\tilde{s} \in \mathcal{S}$ and any $\tilde{v} \in \mathcal{S}^\perp$. By Definition 0.19, $\tilde{s} \cdot \tilde{v} = 0$, so $\tilde{v} \cdot \tilde{s} = 0$ too (Theorem 1 in Vectors). So $\tilde{s}$ is orthogonal to every vector of $\mathcal{S}^\perp$, that is, $\tilde{s} \in (\mathcal{S}^\perp)^\perp$.

Equal dimensions. $\mathcal{S}^\perp$ is a subspace (Theorem 0.29), so part 2 of Theorem 0.31 applies to it as well as to $\mathcal{S}$:

$$\begin{aligned}\dim((\mathcal{S}^\perp)^\perp) &= p - \dim(\mathcal{S}^\perp) && \text{(Theorem 0.31, applied to } \mathcal{S}^\perp\text{)} \\ &= p - (p - \dim(\mathcal{S})) && \text{(Theorem 0.31, applied to } \mathcal{S}\text{)} \\ &= \dim(\mathcal{S}). && \text{(arithmetic)}\end{aligned}$$

$(\mathcal{S}^\perp)^\perp$ is a subspace (Theorem 0.29) that contains $\mathcal{S}$ and has the same dimension, so it equals $\mathcal{S}$ (Theorem 0.11). $\square$

Example 0.57 (Back to the line). In Example 0.56, $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ and $\mathcal{S}^\perp = \text{span}\{(1, -1, 0), (0, 0, 1)\}$. Put those two spanning vectors in the columns of the 3×2 matrix

$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$, so $\mathcal{S}^\perp = \mathcal{C}(\mathbf{B})$ (Theorem 0.12) and $(\mathcal{S}^\perp)^\perp = \mathcal{N}(\mathbf{B}^\top)$ (Theorem 0.30). Since

$\mathbf{B}^\top \tilde{w} = (w_1 - w_2, w_3)$ (Definition 11 in Matrices), $\tilde{w} \in (\mathcal{S}^\perp)^\perp$ exactly when $w_1 - w_2 = 0$ and $w_3 = 0$. So

$$\begin{aligned}(\mathcal{S}^\perp)^\perp &= \{(a, a, 0) : a \in \mathbb{R}\} \\ &= \mathcal{S}.\end{aligned}$$

Example 0.58 (A set that is not a subspace does not come back). The one-element set $\mathcal{X} = \{(1, 1, 0)\}$ is not a subspace. Its orthogonal complement is the same as that of $\mathcal{S}$ in Example 0.57, since both are defined by the single condition $u_1 + u_2 = 0$, so $(\mathcal{X}^\perp)^\perp = \text{span}\{(1, 1, 0)\}$, which contains $(2, 2, 0) \notin \mathcal{X}$. Theorem 0.32 needs $\mathcal{S}$ to be a subspace.

The fundamental theorem of linear algebra

Theorem 0.33 (Fundamental theorem of linear algebra). *Let $\mathbf{A}$ be an $m \times n$ matrix with $\text{rank}(\mathbf{A}) = r$. Then*

1. $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$, and $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$;

2. $\mathcal{C}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\top)^\perp$;

3. $\mathcal{N}(\mathbf{A})^\perp = \mathcal{C}(\mathbf{A}^\top)$, and $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$;

- 4.

$$\begin{aligned} \dim(\mathcal{C}(\mathbf{A})) &= \dim(\mathcal{C}(\mathbf{A}^\top)) \\ &= r, \end{aligned}$$

$$\dim(\mathcal{N}(\mathbf{A})) = n - r, \text{ and } \dim(\mathcal{N}(\mathbf{A}^\top)) = m - r.$$

Proof

Proof. **Part 1.** The first equation is Theorem 0.30. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 0.12), so

$$\begin{aligned} \mathbb{R}^m &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})^\perp && \text{(Theorem 0.31)} \\ &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top). && \text{(first equation)} \end{aligned}$$

Part 2.

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= (\mathcal{C}(\mathbf{A})^\perp)^\perp && \text{(Theorem 0.32)} \\ &= \mathcal{N}(\mathbf{A}^\top)^\perp. && \text{(part 1)} \end{aligned}$$

Part 3. Apply parts 2 and 1 to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$: part 2 gives $\mathcal{C}(\mathbf{A}^\top) = \mathcal{N}(\mathbf{A})^\perp$, and part 1 gives $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.

Part 4. $\dim(\mathcal{C}(\mathbf{A})) = r$ by Theorem 0.16. $\dim(\mathcal{C}(\mathbf{A}^\top)) = \text{rank}(\mathbf{A}^\top)$ by Theorem 0.16, and that is r by Theorem 0.19. $\mathbf{A}$ has n columns, so $\dim(\mathcal{N}(\mathbf{A})) = n - r$ (Definition 0.12, Theorem 0.17). $\mathbf{A}^\top$ has m columns and rank r , so $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$ in the same way. $\square$

Example 0.59 (The four subspaces of the matrix in Example 0.22). For $\mathbf{A}$ in Example 0.22, $m = 2$, $n = 3$ and $r = 1$ (Example 0.29).

- In $\mathbb{R}^3$: the row space $\mathcal{C}(\mathbf{A}^\top) = \text{span}\{(1, -2, -2)\}$
 - (1) has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ has dimension $2 = n - r$ (Example 0.30). The spanning vectors are orthogonal:

$$\begin{aligned}(1, -2, -2) \cdot (2, 1, 0) &= 2 - 2 + 0 \\ &= 0\end{aligned}$$

and

$$\begin{aligned}(1, -2, -2) \cdot (2, 0, 1) &= 2 + 0 - 2 \\ &= 0.\end{aligned}$$

- In $\mathbb{R}^2$: $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$ has dimension $1 = m - r$ (Example 0.55).

By part 3, $\tilde{y} = (1, 0, 0)$ splits uniquely into a row-space part $t(1, -2, -2)$ and a null-space part. The null-space part $\tilde{y} - t(1, -2, -2)$ lies in $\mathcal{N}(\mathbf{A})$ (part 3), and $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ (Theorem 0.30 applied to $\mathbf{A}^\top$, with $(\mathbf{A}^\top)^\top = \mathbf{A}$), so it must be orthogonal to $(1, -2, -2)$:

$$\begin{aligned}0 &= (1, -2, -2) \cdot ((1, 0, 0) - t(1, -2, -2)) && \text{(orthogonality)} \\ &= (1, -2, -2) \cdot (1 \cdot (1, 0, 0) + (-t)(1, -2, -2)) && \text{(write the difference as a linear combination)} \\ &= 1 \cdot ((1, -2, -2) \cdot (1, 0, 0)) + (-t)((1, -2, -2) \cdot (1, -2, -2)) && \text{(Theorem 0.28)} \\ &= 1 \cdot 1 + (-t) \cdot 9 && \text{(Definition 7 in Vectors)} \\ &= 1 - 9t, && \text{(arithmetic)}\end{aligned}$$

so $t = \frac{1}{9}$. The row-space part is $(\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9})$ and the null-space part is $(\frac{8}{9}, \frac{2}{9}, \frac{2}{9})$; as a check, row 1 of $\mathbf{A}$ gives $\frac{8}{9} - 2 \cdot \frac{2}{9} - 2 \cdot \frac{2}{9} = 0$, and row 2 is 3 times row 1, so $\mathbf{A}$ sends the null-space part to $\tilde{0}_2$.

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